



Statistics Department
Arabic Section
Third Level



Econometrics Project

“Analysis of Global Cancer Data (2015–2024)”

Under the supervision of
Dr-Marwah Sabry Siam

Student Name:	ID
Mona Ahmed Abdelmonem	5220205
Reham Mohamed Eldemerdash	5220671
Abdelrahman Ahmed Youssef	5220913
Menna-tallah Mostafa Elsayed	5220819
Salma Khaled Ahmed	5220475

2024-2025

Introduction:

Cancer remains one of the most pressing global health challenges of the 21st century, with millions of new cases diagnosed annually across diverse populations. Understanding the factors that influence cancer severity and patient outcomes is crucial for informing treatment strategies, resource allocation, and public health policy. This project utilizes a comprehensive dataset of cancer patients from multiple countries, covering the period from 2015 to 2024. The data include a wide range of variables, encompassing genetic predispositions, lifestyle factors (such as smoking, alcohol use, and obesity), environmental exposures, and clinical characteristics (including cancer stage and type).

The main objective of this econometric analysis is to explore and quantify the relationships between key clinical, lifestyle, and environmental variables and various cancer-related outcomes. Depending on the focus of each research question, different outcome variables—such as severity scores or survival years—are examined to capture distinct dimensions of the patient experience. By applying appropriate regression models and statistical tests, the project aims to evaluate the significance and interaction of major determinants of cancer burden across a large and diverse global population. The analysis is guided by four targeted research questions, each designed to investigate a specific aspect of the complex relationships underlying cancer outcomes.

Questions:

1. Does the average cancer severity score differ according to obesity level, genetic risk, and cancer stage?
2. To what extent do genetic risk, environmental pollution, and treatment cost influence the severity of cancer cases?
3. Does the average cancer severity score differ by Cancer Stage and Gender, after adjusting for Age and Air Pollution?
4. Is there a statistically significant effect of smoking, alcohol use, obesity, Gender on cancer severity scores, Do the associations between these risk factors and severity scores differ by gender?

Description of the data:

The dataset used in this analysis consists of 33,505 observations and 15 variables, including both numeric and categorical data types. Each row represents a unique cancer patient, capturing demographic, lifestyle, environmental, and clinical features.

The numeric variables:

Age, Genetic Risk, Air Pollution, Alcohol Use, Smoking, Obesity Level, Treatment Cost USD, Survival Years, and Target Severity Score.

The categorical variables:

Gender, Country Region, Cancer Type, and Cancer Stage.

The variable Year is numeric but behaves categorically, indicating the year of diagnosis.

```
> str(data)
tibble [33,505 x 15] (S3: tbl_df/tbl/data.frame)
 $ Age          : num [1:33505] 71 34 80 40 43 22 41 72 21 49 ...
 $ Gender       : chr [1:33505] "Male" "Male" "Male" "Male" ...
 $ Country_Region : chr [1:33505] "UK" "China" "Pakistan" "UK" ...
 $ Year        : num [1:33505] 2021 2021 2023 2015 2017 ...
 $ Genetic_Risk  : num [1:33505] 6.4 1.3 7.4 1.7 5.1 9.5 5.1 6 4.3 8.1 ...
 $ Air_Pollution : num [1:33505] 2.8 4.5 7.9 2.9 2.8 6.4 8.2 8.2 3.8 0.8 ...
 $ Alcohol_Use   : num [1:33505] 9.5 3.7 2.4 4.8 2.3 3.3 0.3 6.4 1 7.8 ...
 $ Smoking       : num [1:33505] 0.9 3.9 4.7 3.5 6.7 3.9 3.7 0.6 0.3 5.2 ...
 $ Obesity_Level : num [1:33505] 8.7 6.3 0.1 2.7 0.5 5.1 2.1 8.5 8.5 9.3 ...
 $ Cancer_Type   : chr [1:33505] "Lung" "Leukemia" "Breast" "Colon" ...
 $ Cancer_Stage  : chr [1:33505] "Stage III" "Stage 0" "Stage II" "Stage I" ...
 $ Treatment_Cost_USD : num [1:33505] 62913 12573 6984 67446 77977 ...
 $ Survival_Years : num [1:33505] 4.246 4.368 4.407 0.815 1.32 ...
 $ Target_Severity_Score: num [1:33505] 7.47 3.99 5.81 5.74 6.33 ...
```

The numeric variables:

1. Age ranges from 20 to 89 years, with a mean and median around 54, suggesting a slightly older patient population overall. The age distribution is fairly symmetrical based on the equal spacing of quartiles.
2. Treatment Cost USD spans from \$5,000 to \$100,000, with a median of ~\$52,000, and a slight right skew, likely due to high-cost cases.
3. Survival Years ranges from 0.5 to 9.95 years, with a mean of 3.53 and median of 3.14, suggesting many patients survive multiple years post-diagnosis.
4. Target Severity Score, the central outcome metric, is normally distributed with a mean of 6.2 and IQR of ~5.0 to 7.5, indicating moderate to high severity in most patients.
5. Each of the following is measured on a 0–10 scale, reflecting standardized intensities or exposure levels: (Genetic Risk, Air Pollution, Alcohol Use, Smoking, and Obesity Level) all show:
 - **Mean $\approx$ Median $\approx$ 5.**
 - **Interquartile range: 2.5 to 7.5,”** Indicating symmetrical and balanced distributions”.

The categorical variables:

1. Cancer Stage Distribution:

The dataset includes patients across all cancer stages: Stage II is the most common (6,796 cases), followed closely by Stage I (6,737) and Stage IV (6,708). The distribution is fairly balanced across all five stages, suggesting a diverse patient population from early detection to advanced stages. The near-equal representation supports comparative analysis across stages.

```
> # Frequency of Cancer Stage
> table(data$Cancer_Stage)
```

Stage 0	Stage I	Stage II	Stage III	Stage IV
6620	6737	6796	6644	6708

2. Cancer Type Distribution:

The types of cancer are also evenly distributed: Each cancer type (e.g., Breast, Colon, Lung, Prostate) has ~4,100–4,250 cases, with Colon cancer being slightly most frequent (4,255). This balance indicates that the dataset is well-suited for multi-cancer comparative studies, avoiding bias from overrepresentation of a single type.

```
> # Frequency of Cancer Type
> table(data$Cancer_Type)
```

Breast	Cervical	Colon	Leukemia	Liver	Lung	Prostate	Skin
4131	4188	4255	4173	4155	4170	4251	4182

3. Gender Distribution:

Males and females are nearly equally represented, with 16,796 males and 16,709 females, this parity provides a solid basis for gender-based analysis of cancer outcomes, treatment costs, and risk profiles.

```
> # Frequency of gender
> table(data$Gender)
```

Female	Male
16709	16796

4. Geographic Representation:

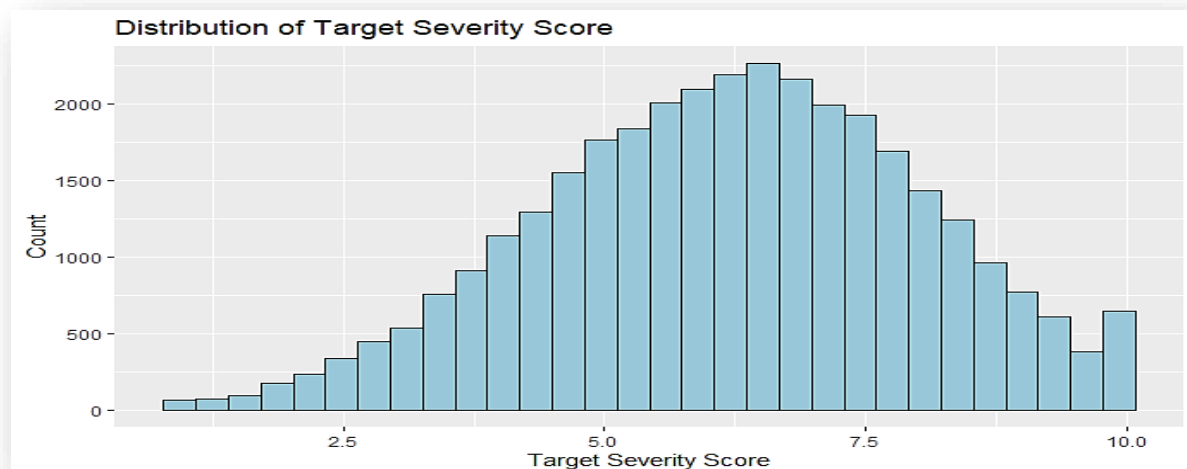
The dataset covers patients from 10 countries, with each contributing approximately 3,250 to 3,450 cases, UK (3,411) and Australia (3,454) have slightly higher frequencies, but the differences are minor, this even distribution supports cross-country comparisons and minimizes location bias.

```
> # Frequency of Country Region  
> table(data$Country_Region)
```

Australia	Brazil	Canada	China	Germany	India	Pakistan	Russia	UK	USA
3454	3353	3252	3305	3341	3357	3310	3345	3411	3377

Distribution of Target Severity Score:

The distribution of the Target Severity Score appears to be approximately bell-shaped and slightly left-skewed, suggesting a near-normal distribution with a slight concentration of lower severity scores. Most patients have scores ranging from 4 to 8, with a peak around 6 to 7, indicating that moderate to high severity cases are the most common. There is a smaller but noticeable proportion of patients with extremely low (close to 1) or extremely high (close to 10) severity scores.



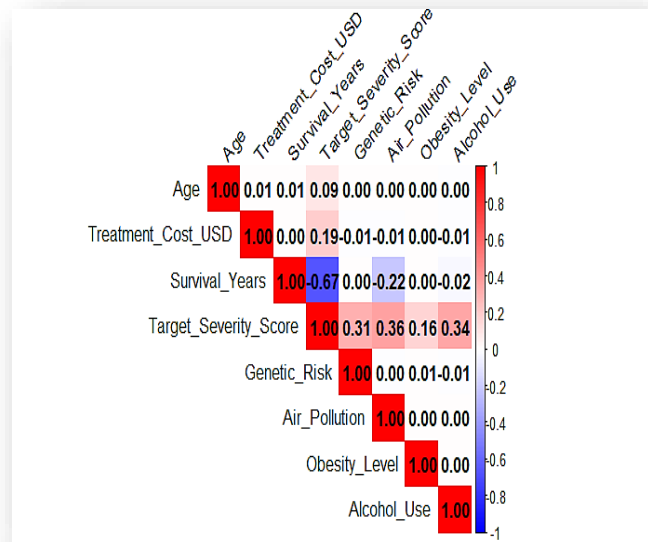
Correlation Matrix Analysis:

Survival Years and Target Severity Score exhibit a moderate negative correlation of approximately -0.67, indicating that higher severity scores are associated with shorter survival times.

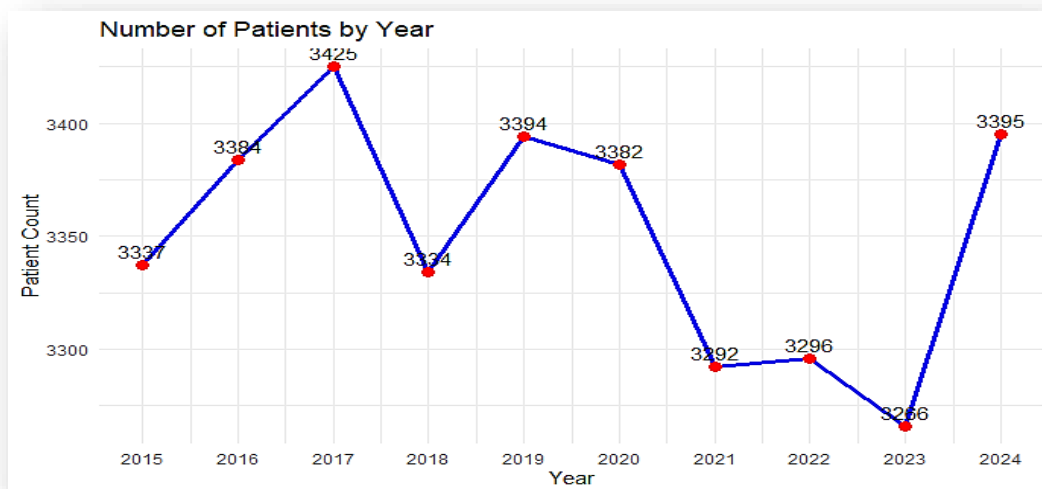
. Target Severity Score shows positive but weak correlations with:

- Genetic Risk (0.31),
- Obesity Level (0.16),
- Alcohol Use (0.34).

” Suggesting that lifestyle and genetic factors may contribute to the worsening of cancer severity”.



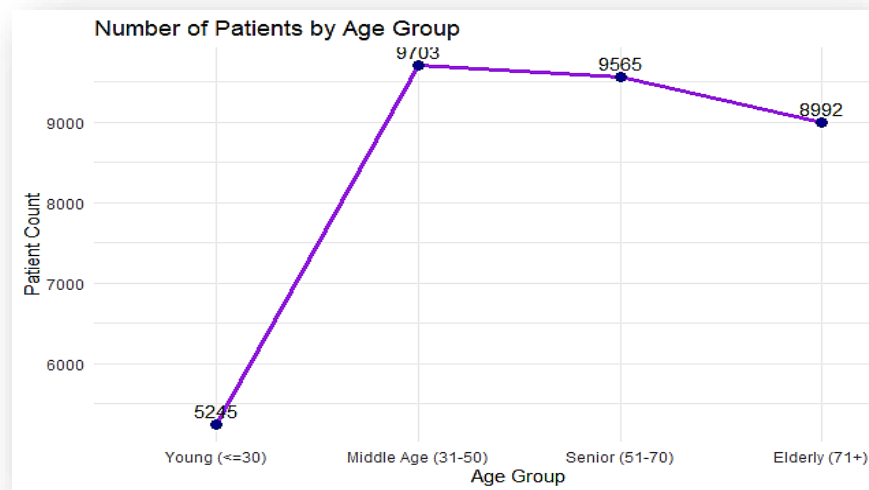
Temporal Distribution of Patients (2015–2024):



The dataset spans ten years and displays mild fluctuations in the number of patients per year. The highest patient count occurred in 2017 (3,425 patients), followed by a notable dip in 2018 (3,344 patients). A second local peak is seen in 2019 (3,394), after which the numbers gradually decline, reaching a minimum of 3,266 patients in 2023. Interestingly, there is a sharp rebound in 2024 (3,395 patients), these fluctuations may reflect variations in data collection efforts, access to healthcare services, or external factors such as the COVID-19 pandemic around 2020–2022, which may have affected diagnosis and reporting rates.

Number of Patients by Age Group:

The chart illustrates the distribution of cancer patients by age group. The Middle Age group (31–50) comprises the largest number of patients (9,703), followed closely by the Senior group (51–70) with 9,565 patients. The Elderly group (71+) also represents a significant portion of the dataset, with 8,992 patients. In contrast, the young group (≤ 30) accounts for the fewest patients, only 5,245. This pattern suggests that cancer incidence is most concentrated among middle-aged and older adults, which is consistent with epidemiological expectations that cancer risk increases with age. However, the notably high patient count in the 31–50 age group may warrant further investigation, potentially reflecting earlier onset or improved diagnostic practices in this demographic.



Cleaning the data:

1. Missing values per column:

The analysis of missing values reveals that there are no missing entries in any of the variables.

```
> # Count of missing values per column
> colsums(is.na(data))
```

Age	Gender	Country_Region	Year
0	0	0	0
Genetic_Risk	Air_Pollution	Alcohol_Use	Smoking
0	0	0	0
Obesity_Level	Cancer_Type	Cancer_Stage	Treatment_Cost_USD
0	0	0	0
Survival_Years	Target_Severity_Score	Age_Group	
0	0	0	

2. Outliers' detection process:

Outlier detection was conducted on all numeric variables using the IQR method. The results indicate that:

- No outliers were detected in the variables: Age, Year, Genetic Risk, Air Pollution, Alcohol Use, Smoking, Obesity Level, Treatment Cost USD, and Survival Years.
- 69 outliers were identified in Target Severity Score.

To avoid these extreme values from affecting our results, we removed them from the dataset. This gave us a cleaned version of the data, which we used for our analysis to make the results more accurate and reliable.

```
> outlier_counts
      Age      Year      Genetic_Risk      Air_Pollution
      0         0         0             0
Alcohol_Use      Smoking      Obesity_Level      Treatment_Cost_USD
      0         0         0             0
Survival_Years      Target_Severity_Score
      0         69
```


Question 1: Does the average cancer severity score differ by Cancer Stage and Gender, after adjusting for Age and Air Pollution?

The question asks whether cancer severity differs by cancer stage and gender, after adjusting for age and air pollution. This is important because it:

- Improves clinical decisions by identifying who is most at risk.
- Reveals possible biological or gender-based differences.
- Helps shape public health policies (e.g., screening vs. pollution control).
- Supports health equity by identifying vulnerable groups.
- Ensures statistical accuracy by removing confounding variables.
- Guides better research design and future studies.

The explanatory variables include both quantitative and qualitative variables. Regression models containing mixture of quantitative and qualitative variables are called analysis of covariance (ANCOVA) models.

So, to answer this question we must use ANCOVA model because we have quantitative variables like Age and Air Pollution and qualitative variables like Cancer Stage and Gender.

The structure of the question:

- Dependent Variable: "Average cancer severity score" (this is a continuous variable).
- Independent Variables (Factors): "Cancer Stage" (categorical) and "Gender" (categorical).
- Covariates (Variables to Control For): "Age" (continuous) and "Air Pollution" (continuous).

Covariates are continuous variables that we include in a model to control for their effect on the dependent variable.

- They are not the main focus (unlike your categorical factors) but can influence the outcome.
- By including covariates, we adjust the comparison between groups to account for differences in these variables.

Possible models:

Let:

- Target severity score: Y_i
- Age: X_{1i}
- Air Pollution: X_{2i}
- $D_{1i} = \begin{cases} 1 & \text{patient is in stage I or II} \\ 0 & \text{O.W} \end{cases}$

- $D_{2i} = \begin{cases} 1 & \text{patient is in stage III or VI} \\ 0 & \text{O.W} \end{cases}$
- $D_{3i} = \begin{cases} 1 & \text{patient is male} \\ 0 & \text{O.W} \end{cases}$

Model without interaction terms:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_{1i} + \beta_5 X_{2i} + u_i$$

Models with interaction terms:

Model (2):

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 (D_{1i} \cdot D_{3i}) + \beta_5 (D_{2i} \cdot D_{3i}) + \beta_6 X_{1i} + \beta_7 X_{2i} + u_i$$

Model (3):

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_{1i} + \beta_5 X_{2i} + \beta_6 (D_{3i} \times X_{1i}) + \epsilon_i$$

Model (4):

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_{2i} + \beta_5 X_{1i} + \beta_6 (D_{3i} X_{2i}) + \epsilon_i$$

Full model:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_{1i} + \beta_5 X_{2i} + \beta_6 (D_{1i} \cdot D_{3i}) + \beta_7 (D_{2i} \cdot D_{3i}) + \beta_8 (D_{1i} \cdot X_{2i}) + \beta_9 (D_{2i} \cdot X_{2i}) + \beta_{10} (D_{3i} \cdot X_{2i}) + \dots + u_i$$

- Model without interaction terms:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_{1i} + \beta_5 X_{2i} + u_i$$

- Mean severity score for a female patient in stage 0:

$$E(Y_i | D_{1i} = 0, D_{2i} = 0, D_{3i} = 0) = \beta_0 + \beta_4 X_{1i} + \beta_5 X_{2i}$$

- Mean severity score for a female patient in stage I or II:

$$E(Y_i | D_{1i} = 1, D_{2i} = 0, D_{3i} = 0) = (\beta_0 + \beta_1) + \beta_4 X_{1i} + \beta_5 X_{2i}$$

- Mean severity score for a female patient in stage III or IV:

$$E(Y_i | D_{1i} = 0, D_{2i} = 1, D_{3i} = 0) = (\beta_0 + \beta_2) + \beta_4 X_{1i} + \beta_5 X_{2i}$$

- Mean severity score for a male patient in stage 0:

$$E(Y_i | D_{1i} = 0, D_{2i} = 0, D_{3i} = 1) = (\beta_0 + \beta_3) + \beta_4 X_{1i} + \beta_5 X_{2i}$$

- Mean severity score for a male patient in stage I or II:

$$E(Y_i | D_{1i} = 1, D_{2i} = 0, D_{3i} = 1) = (\beta_0 + \beta_1 + \beta_3) + \beta_4 X_{1i} + \beta_5 X_{2i}$$

- Mean severity score for a male patient in stage III or IV:

$$E(Y_i | D_{1i} = 0, D_{2i} = 1, D_{3i} = 1) = (\beta_0 + \beta_2 + \beta_3) + \beta_4 X_{1i} + \beta_5 X_{2i}$$

```
Call:
lm(formula = Target_Severity_Score ~ Cancer_Stage_Grouped + Gender +
    Age + Air_Pollution, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-5.5821 -1.1184  0.0347  1.1592  5.2331

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      3.8886354   0.0360621  107.83  <2e-16 ***
Cancer_Stage_Groupedstage_1_2  0.3942237   0.0244470   16.13  <2e-16 ***
Cancer_Stage_Groupedstage_3_4  0.7988476   0.0245001   32.61  <2e-16 ***
GenderMale        0.5130296   0.0178084   28.81  <2e-16 ***
Age               0.0080215   0.0004418   18.16  <2e-16 ***
Air_Pollution    0.2272445   0.0030816   73.74  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.63 on 33499 degrees of freedom
Multiple R-squared:  0.1871,    Adjusted R-squared:  0.1869
F-statistic: 1542 on 5 and 33499 DF,  p-value: < 2.2e-16
```

- Residuals:** The residual statistics suggest a reasonable fit for the model, with predictions generally close to observed values and a distribution centered around zero. There might be a slight tendency for larger underpredictions compared to overpredictions, but overall, the model seems to perform adequately based on these statistics alone.
- The interpretation of coefficients:** all the coefficients are significant as $p\text{-value} = 2e-16 < 0.05$ so we will reject the null hypothesis which says for each coefficient that it is equal to 0.
 - β_0 : the average value of target severity score for female in stage 0 will be 3.8886354 when age and air pollution at their lowest values.
 - β_1 : the difference in the average value in the target severity score between the patient who in stage I or II and stage 0 will be 0.3942237, holding all other explanatory variables constant.
 - β_2 : the difference in the average value in the target severity score between the patient who in stage III or VI and stage 0 will be 0.7988476, holding all other explanatory variables constant.

- β_3 : the difference in the average value in the target severity score between a male patient and a female patient will be 0.5130296, holding all other explanatory variables constant.
- β_4 : when age increases by 1 year the average value of the target severity score will increase by 0.0080215, holding all other variables constant.
- β_5 : when the air pollution increases by 1 unit the average value of the target severity score will increase by 0.2272445, holding all other variables constant.

3. The analysis shows a statistically significant regression model, and the R-squared value of 0.1871 is relatively low, meaning that the model only explains about 18.7% of the variance in the dependent variable. While the model is statistically robust in identifying relationships, its practical explanatory power is limited. This suggests that while the predictors do have an effect, there are many other factors not included in this model that influence the dependent variable.

- Model with interaction terms:

There are many possible models so we will choose the one we are going to work with based on Anova test and AIC.

The ANOVA test:

```
> anova(model2, model3, model4)
Analysis of Variance Table

Model 1: Target_Severity_Score ~ Cancer_Stage * Gender + Age + Air_Pollution
Model 2: Target_Severity_Score ~ Cancer_Stage + Gender * Age + Air_Pollution
Model 3: Target_Severity_Score ~ Cancer_Stage_Grouped + Gender * Air_Pollution +
      Age
  Res.Df  RSS Df Sum of Sq    F    Pr(>F)
1  33493 88667
2  33496 87825 -3      841.3
3  33498 88970 -2   -1144.8 216.22 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model (4) is statistically significant in explaining a portion of Target severity score variance, driven by the strong relationships with Cancer stage and Air pollution, and supported by an extremely large sample size. The model also benefits from no multicollinearity and no autocorrelation, which are excellent diagnostic results.

According to AIC:

The AIC shows that the best model is model (4) because it has the lowest value and the lowest degree of freedom.

We did not add the full into our comparison because by default it will be the best model because it has the most interaction variables so it will be biased to add it.

	df	AIC
model2	13	127715.7
model3	10	127390.3
model4	8	127820.2

The model:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + \beta_4 X_{2i} + \beta_5 X_{1i} + \beta_6 (D_{3i} \times X_{2i}) + \epsilon_i$$

- Mean severity score for a female patient in stage 0:

$$E[Y_i] = \beta_0 + \beta_4 X_{2i} + \beta_5 X_{1i}$$

- Mean severity score for a female patient in stage I or II:

$$E[Y_i] = \beta_0 + \beta_1 + \beta_4 X_{2i} + \beta_5 X_{1i}$$

- Mean severity score for a female patient in stage III or IV:

$$E[Y_i] = \beta_0 + \beta_2 + \beta_4 X_{2i} + \beta_5 X_{1i}$$

- Mean severity score for a male patient in stage 0:

$$E[Y_i] = \beta_0 + \beta_3 + \beta_4 X_{2i} + \beta_6 X_{2i} + \beta_5 X_{1i}$$

- Mean severity score for a male patient in stage I or II:

$$E[Y_i] = \beta_0 + \beta_1 + \beta_3 + \beta_4 X_{2i} + \beta_6 X_{2i} + \beta_5 X_{1i}$$

- Mean severity score for a male patient in stage III or IV:

$$E[Y_i] = \beta_0 + \beta_2 + \beta_3 + \beta_4 X_{2i} + \beta_6 X_{2i} + \beta_5 X_{1i}$$

```

Residuals:
    min       1q   median       3q      max
-5.5573 -1.1194  0.0356  1.1609  5.2489

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.8616260   0.0391675   98.593  <2e-16 ***
Cancer_Stage_GroupedStage_1_2  0.3941113   0.0244463   16.122  <2e-16 ***
Cancer_Stage_GroupedStage_3_4  0.7987120   0.0244994   32.601  <2e-16 ***
GenderMale      0.5676853   0.0356947   15.904  <2e-16 ***
Air_Pollution  0.2326811   0.0043548   53.431  <2e-16 ***
Age            0.0080178   0.0004418   18.148  <2e-16 ***
GenderMale:Air_Pollution -0.0108879   0.0061626   -1.767   0.0773 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.63 on 33498 degrees of freedom
Multiple R-squared:  0.1871,    Adjusted R-squared:  0.187
F-statistic: 1285 on 6 and 33498 DF,  p-value: < 2.2e-16

```

1. **Residuals:** The residuals show reasonable symmetry but notable extremes. Their range should be -3:3 but we can ignore them as the sample size is so large
2. **The interpretation of coefficients:** all the coefficients are significant as $p\text{-value} = 2e-16 < 0.05$ so we will reject the null hypothesis which says for each coefficient that it is equal to 0 except for the interaction term as we do reject the null hypothesis at 0.1.
 - β_0 : the average value of target severity score for a female patient in stage 0 will be 3.8616260, when air pollution and age at their lowest value.
 - β_1 : the difference in the average value of target severity score for a patient in stage I or II and stage 0 will be 0.3941113, holding all other variables constant.
 - β_2 : the difference in the average value target severity score patient in stage III or VI and stage 0 will be 0.7987120, holding all other variables constant.
 - β_3 : the difference in the average value target severity score for a male patient and a female patient will be 0.5676853, when air pollution equal to 0 and other variables are constant.
 - β_4 : when the air pollution increases by 1 unit the average value of target severity score will increase by 0.2326811 for female patients and other variables are constant.
 - β_5 : when the age increases by 1 year the average value of target severity score will increase by 0.0080178, holding all other variables constant.
 - β_6 : when air pollution increases by 1 unit the difference average value of target severity score will decrease by 0.0108879 between male patients and female patients and other variables are constant.
3. The model is statistically significant (indicated by the high F-statistics and extremely low p-value), meaning that the predictors, as a group, have a significant relationship with the dependent variable. However, the low R-squared value (0.1871) suggests that while statistically significant, the model only explains a small proportion of the variance in the dependent variable. This implies that while the chosen predictors are relevant, a substantial amount of the variability in the dependent variable remains unexplained.

Now we are going to compare the model with interaction term and the model without interaction term to choose the best model and check its assumptions.

The Anova test:

```
> anova( model11 ,model14 )
Analysis of Variance Table

Model 1: Target_Severity_Score ~ Cancer_Stage_Grouped + Gender + Age +
  Air_Pollution
Model 2: Target_Severity_Score ~ Cancer_Stage_Grouped + Gender * Air_Pollution +
  Age
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1   33499 88978
2   33498 88970   1     8.2906 3.1215 0.07727 .
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The ANOVA results suggest that including an interaction between Gender and Air pollution does not significantly improve the model's ability to predict Target severity score compared to a model with only additive effects, at a standard 0.05 significance level.

Therefore, Model (1) is the preferred choice because it achieves a comparable fit without the added complexity of an interaction term that isn't statistically significant.

According to AIC:

While model4 has a slightly lower AIC, ANOVA explicitly tested the statistical significance of the extra term in model (4) (the interaction term), and it found that the term was not statistically significant at the common $\alpha=0.05$ level ($p\text{-value} = 0.07727$).

	df	AIC
model1	7	127821.3
model4	8	127820.2

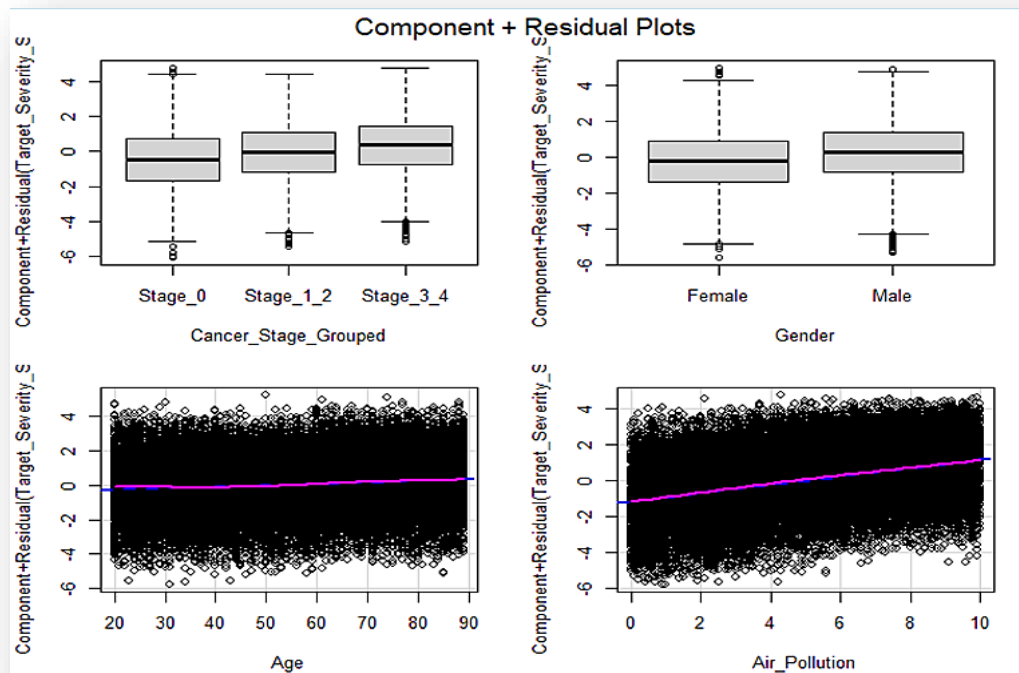
Therefore, the most robust recommendation, considering both the statistical significance test and the information criterion, remains to choose model (1).

The marginal difference in AIC is not enough to override the lack of statistical significance for the additional parameter in model (4). Sticking with model (1) aligns with the principle of parsimony and avoids including terms that do not demonstrably improve the model's fit in a statistically significant way.

So, we are going to check the assumptions of model (1).

Checking the assumptions:

1. Linearity:



- Cancer stage appears to have a positive association with Target severity score, with higher stages tending to have higher severity.
- Gender does not seem to have a significant unique relationship with Target severity score.
- Age shows very little or no linear relationship with Target severity score when other predictors are in the model.
- Air pollution exhibits a clear positive linear relationship with Target severity score, suggesting that higher air pollution is associated with increased target severity.

2. Independence of errors:

```
> durbinwatsonTest(model1)
lag Autocorrelation D-W Statistic p-value
1 0.003776783 1.992299 0.506
Alternative hypothesis: rho != 0
```

- The null hypothesis: there is no autocorrelation
- The alternative hypothesis: there is autocorrelation

p-value=0.516>0.05 so we will not reject the null hypothesis which means there is no autocorrelation.

This is a desirable outcome for a regression model, as it indicates that the assumption of independent errors is likely met. This helps ensure that the standard errors of the regression coefficients are unbiased and that the p-values for individual predictors are reliable.

3. Multicollinearity:

```
> vif(model1)
          GVIF Df GVIF^(1/(2*Df))
Cancer_Stage_Grouped 1.000268 2 1.000067
Gender 1.000102 1 1.000051
Age 1.000071 1 1.000035
Air_Pollution 1.000197 1 1.000098
```

The VIF values for all predictors in the model are extremely close to 1. This is an excellent result and indicates that there is virtually no multicollinearity among the predictor variables (Cancer stage, Gender, Age, and Air pollution) in this regression model.

And indicates the model variables are not linearly dependent and the coefficient estimates are likely to be stable and reliable.

4. Homoscedasticity:

```
studentized Breusch-Pagan test  
  
data: model1  
BP = 280.37, df = 5, p-value < 2.2e-16
```

The null hypothesis of the Breusch-Pagan test is that homoscedasticity is present (constant variance of residuals).

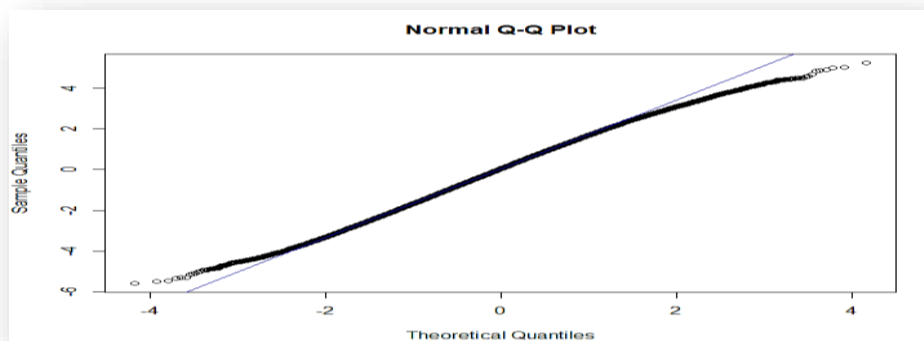
The very small p-value ($< 2.2e-16$) means we strongly reject the null hypothesis.

Therefore, there is significant evidence of heteroscedasticity in the model.

We could solve this problem by using the following code:

```
#Robust standard errors (HC3 is a common choice)  
install.packages("sandwich")  
library(sandwich)  
library(lmtest)  
coefTest(model1, vcov = vcovHC(model1, type = "HC3"))  
# Install if needed  
install.packages("estimatr")  
# Load the package  
library(estimatr)  
# Fit model with robust SEs  
model1_robust <- lm_robust(Target_Severity_Score ~ Cancer_Stage_Grouped + Gender + Age + Air_Pollution, data = data)  
# Summary  
summary(model1_robust)  
plot(fitted(model1), residuals(model1))  
abline(h = 0, col = "red")
```

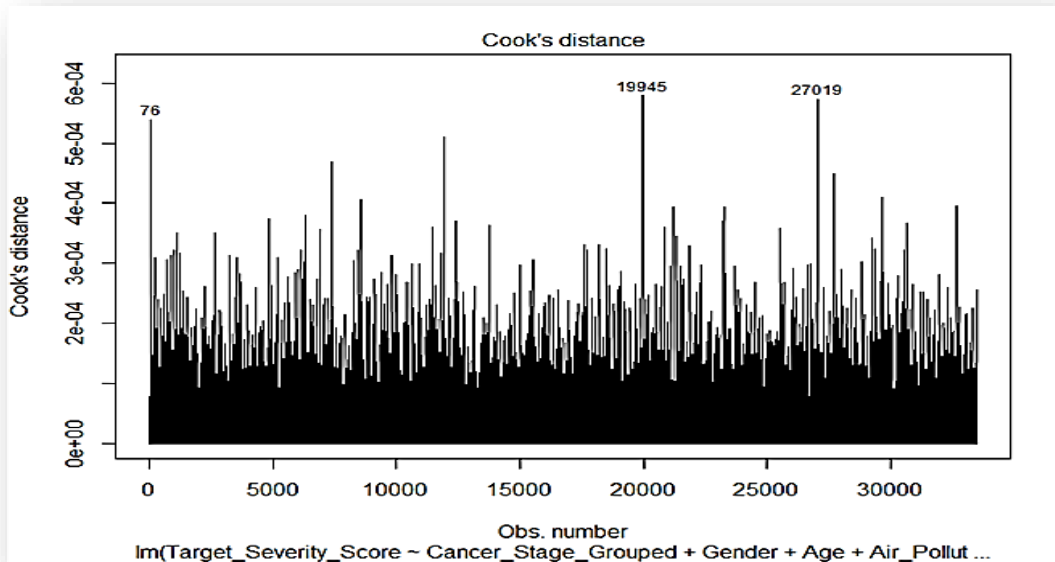
5. Normality (using QQ plot):



The Normal Q-Q plot suggests that while the bulk of the data (residuals) are reasonably normally distributed, there are deviations from normality in both the lower and upper tails. Specifically, the tails of the distribution are "heavier" or "fatter" than those of a true normal distribution. This means

we have more extreme (both positive and negative) residuals than would be expected if the residuals were perfectly normally distributed.

6. Influential observations:



The Cook's distance plot for the model identifies a few influential observations (76, 19945, 27019). While their Cook's distances are higher than the majority, they are still relatively small in absolute terms given the large sample size.

- **Conclusion:**

The average cancer severity score indeed differs significantly by Cancer Stage and Gender, even after adjusting for Age and Air Pollution. Additionally, both Age and Air Pollution show significant individual effects on severity. Model (1), the additive model, is the most appropriate choice given the lack of statistically significant interaction effects. While the model is statistically robust and the coefficients are reliable (due to low multicollinearity and no autocorrelation), the presence of heteroscedasticity is a limitation that needs to be addressed (using robust standard errors). The relatively low R-squared value also highlights that other unmeasured factors play a substantial role in explaining cancer severity.

Question 2: To what extent do genetic risk, environmental pollution, and treatment cost influence the severity of cancer cases?

while many cancer risk factors are linked to lifestyle choices, others stem from external forces beyond an individual's control. Understanding how non-modifiable factors contribute to cancer outcomes is essential for developing broader public health strategies and healthcare system interventions. In particular, genetic predisposition, environmental pollution, and treatment cost are three variables that are often outside the patient's personal agency, yet may play a substantial role in disease progression and patient survival.

This question focuses on assessing the relationship between three such factors Genetic Risk, Air Pollution, and Treatment Cost in USD and their potential impact on cancer outcomes. These variables were selected deliberately due to their nature as uncontrollable or difficult-to-modify influences. For example, genetic risk is inherited and largely unavoidable, air pollution is an environmental condition often determined by geographic and socio-economic factors, and treatment cost is tied to systemic healthcare structures.

By using real-world patient data, we aim to evaluate how these external factors correlate with cancer severity. The analysis will be conducted using linear regression models in R.

- **Model:**

In this case we will try the “Multiple Linear Regression”, the model will be $(\text{Target_Severity_Score}_i = \beta_0 + \beta_1 \cdot \text{Genetic_Risk}_i + \beta_2 \cdot \text{Air_Pollution}_i + \beta_3 \cdot \text{Treatment_Cost_USD}_i + \epsilon_i)$. We will use R to fit this model as following:

```
Call:
lm(formula = Target_Severity_Score ~ Genetic_Risk + Air_Pollution +
    log(Treatment_Cost_USD), data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-5.7656 -1.0773  0.0574  1.1185  4.9212

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.909681   0.130892   -6.95 3.72e-12 ***
Genetic_Risk   0.194597   0.002954   65.88 < 2e-16 ***
Air_Pollution  0.228179   0.002940   77.61 < 2e-16 ***
log(Treatment_Cost_USD) 0.467837   0.012069   38.76 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.555 on 33501 degrees of freedom
Multiple R-squared:  0.2598,    Adjusted R-squared:  0.2598
F-statistic: 3920 on 3 and 33501 DF, p-value: < 2.2e-16
```

- Interpretation of Coefficients:

Predictor	Coefficient (Estimate)	p-value	Interpretation
Intercept	-0.909681	3.72e-12	Statistically significant at $\alpha=0.05$. The expected severity score when all predictors are zero is “-0.909681” (not practically interpretable but necessary for model calculation).
Genetic_Risk	0.194597	< 2e-16	Statistically significant at $\alpha=0.05$. when Genetic_Risk increases by 1 unit the severity score will increase by “0.194597”, on average.
Air_Pollution	0.228179	< 2e-16	Statistically significant at $\alpha=0.05$. when Air_Pollution increases by 1 unit the severity score will increase by “0.228179”, on average.
log(Treatment_Cost_USD)	0.467837	< 2e-16	Statistically significant. A 1% increase in treatment cost is associated with a “0.467837” increase in severity score.

Notes:

1. The intercept in this regression model is negative (-0.91), indicating the expected cancer severity score when all predictors (genetic risk, air pollution, and log treatment cost) are zero, which is in most real-world situations (especially in health and social sciences) having all predictors equal to zero is either unrealistic or outside the range of observed data.
2. Since the Treatment_Cost_USD in our dataset is a large number, we take the log transformation to it to make the analysis more realistic because if we don't take the transformation, it will affect the value of the coefficient's as following:

```

Residuals:
    Min       1Q   Median       3Q      Max
-5.6776 -1.0708  0.0573  1.1195  4.7841

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.403e+00  2.784e-02  122.22  <2e-16 ***
Genetic_Risk  1.947e-01  2.943e-03   66.15  <2e-16 ***
Air_Pollution 2.282e-01  2.930e-03   77.87  <2e-16 ***
Treatment_Cost_USD 1.296e-05  3.103e-07   41.78  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.55 on 33501 degrees of freedom
Multiple R-squared:  0.2649,    Adjusted R-squared:  0.2649
F-statistic: 4025 on 3 and 33501 DF,  p-value: < 2.2e-16

> |

```

So, we will use the model with log transformation of (Treatment_Cost_USD)

- **Model Fit:**

1. **Multiple R-squared = 0.2598**
2. **Adjusted R-squared = 0.2598**

This means that approximately 26% of the variation in cancer severity is explained by the model. While not high, this is reasonable in health-related models where many unobserved factors may exist.

- **Overall Model Significance:**

1. F-statistic = 3920 (on 3 and 33501 degrees of freedom).
2. p-value < 2.2e-16.

This indicates the model is statistically significant overall. When we take the predictors together, reliably explain variation.

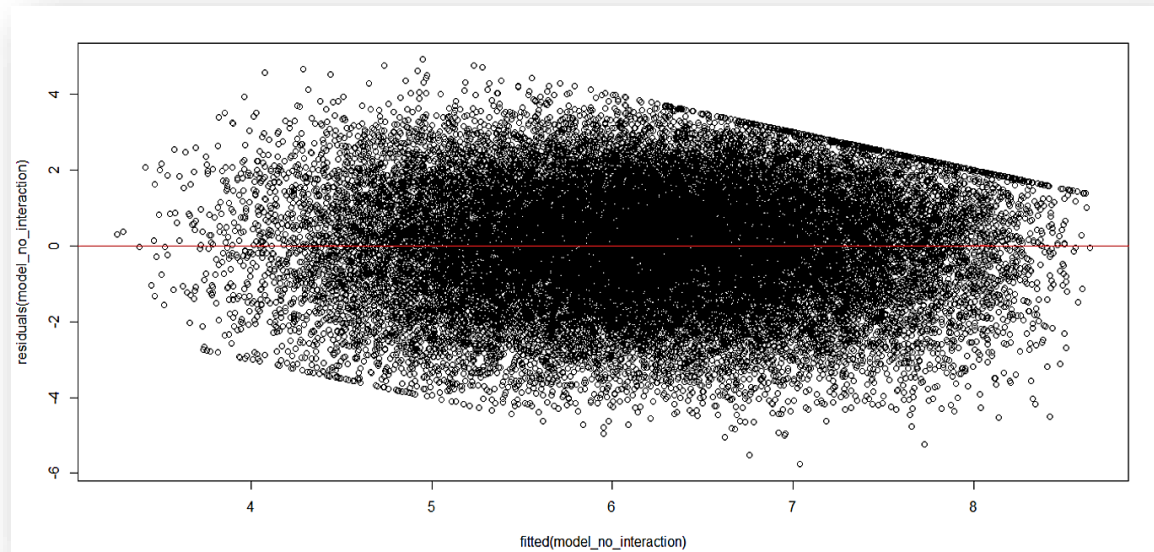
- **Residuals:**

The residuals range from approximately -5.77 to 4.92 which is acceptable. The median residual is close to zero, suggesting that the model is not biased.

- **Checking assumptions:**

1. **Linearity:**

We will check the linearity of the residuals as it is a main assumption.



The residual vs fitted plot shows that residuals are centered around zero, suggesting no major bias in predictions. We can say that the variance is somewhat constant so we can say that the relation is linear.

2. Autocorrelation:

We will check the autocorrelation of the residuals using Durbin-Watson test which its hypothesis is:

$H_0: \rho=0$ “There is no autocorrelation among the residuals.”

$H_1: \rho \neq 0$ “There is autocorrelation among the residuals.”

```
Durbin-Watson test  
  
data: model_no_interaction  
DW = 1.9928, p-value = 0.2552  
alternative hypothesis: true autocorrelation is greater than 0
```

Since the p-value > 0.05 so we will not reject H_0 then there is no autocorrelation among the residuals.

3. multicollinearity:

we will check it using VIF:

```
Genetic_Risk      Air_Pollution log(Treatment_Cost_USD)  
1.000044          1.000054          1.000057  
>
```

All variables have VIF values very close to 1, indicating that there is no multicollinearity between the predictors in the model. This satisfies one of the key assumptions of multiple linear regression, and suggests that each predictor contributes unique information to the model without redundancy.

4. Heteroscedasticity:

We will check it using Breusch-Pagan test. The hypothesis of this test is as following:

H_0 : The variance of the residuals is constant (i.e., **homoskedasticity**)” $\text{Var}(\epsilon_i) = \sigma^2$ ”.

H_1 : The variance of the residuals is not constant (i.e., **heteroskedasticity**)” $\text{Var}(\epsilon_i) = \sigma^2 h(Z_i)$ ”.

```
studentized Breusch-Pagan test  
  
data: model_no_interaction  
BP = 61.859, df = 3, p-value = 2.355e-13  
>
```

Since the p-value is less than 0.05 then we will reject H_0 (The variance of the residuals is not constant” **heteroskedasticity**”). We will solve this problem using the following code:

```
# Install required packages (only run once)
install.packages("lmtest")
install.packages("sandwich")

# Load packages
library(lmtest)
library(sandwich)

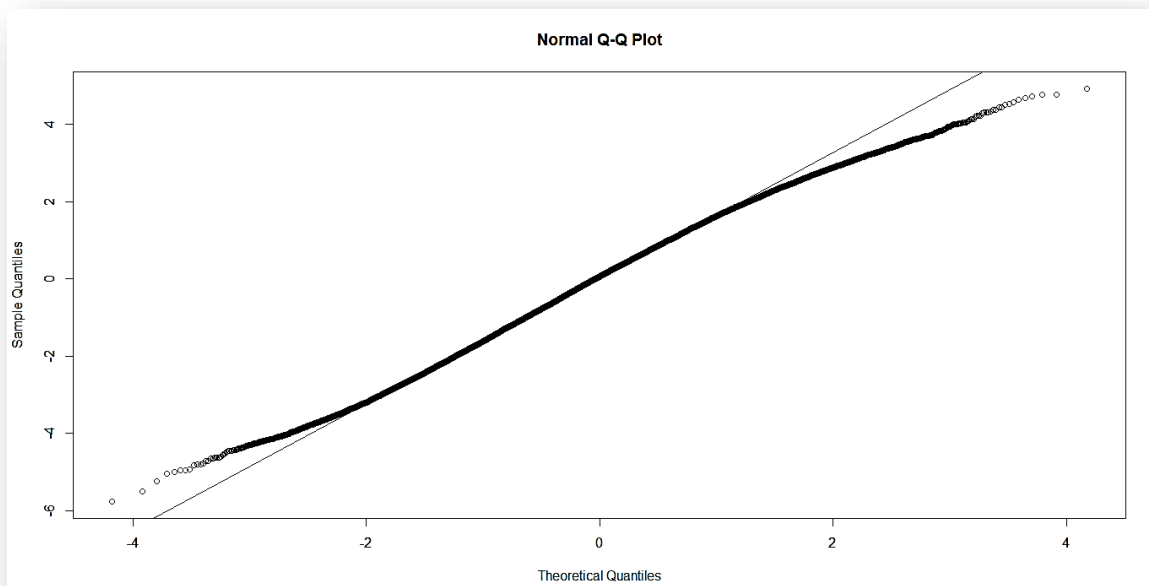
# Fit the OLS model
model <- lm(Target_Severity_Score ~ Genetic_Risk + Air_Pollution + Treatment_Cost_USD, data = your_data)

# Run Breusch-Pagan test for heteroskedasticity
bp_result <- bptest(model)
print(bp_result)

# Apply robust standard errors (HC1 = robust, like in Stata)
robust_results <- coeftest(model, vcov = vcovHC(model, type = "HC1"))
print(robust_results)
```

5. Normality:

We will check normality of residuals using the Q-Q plot:



Overall Shape: The points generally follow a straight line with small deviations at the tails, suggesting that the data approximately follows a normal distribution.

- **Conclusion:**

This analysis shows that genetic risk, air pollution, and treatment cost all have a clear effect on how severe a person's cancer is. People with higher genetic risk, who live in more polluted areas, or who have higher treatment costs tend to have more serious cases. Although the model doesn't explain all the reasons behind cancer severity, it still gives useful information. The results are reliable because the data met most of the important statistical checks. Overall, this study shows that not just personal choices, but also factors beyond a person's control like their genes, environment, and financial situation can affect how serious their illness becomes.

Question 3: Does the average cancer severity score differ according to obesity level, genetic risk, and cancer stage?

- **Description of Variables Used in the Model**

In this analysis, a multiple linear regression model was applied to predict the value of a dependent variable based on several independent variables:

Variable	Type	Description
Target Severity Score	Numeric (Dependent Y)	A measure reflecting the severity score of cancer diagnosis.
Obesity Level	Numeric (Independent)	Represents the patient's obesity level, ranging from 0 to 10.
Cancer Stage	Categorical (Independent)	The cancer stage (Stage 0, I, II, III, IV).
Genetic Risk	Numeric (Independent)	Genetic risk score for cancer, ranging from 0 to 10.

- **Model Specification:**

The estimated regression equation is as follows:

Models without interaction terms: Model (1):

$$Y_i = \beta_0 + \beta_1 D1_i + \beta_2 D2_i + \beta_3 D3_i + \beta_4 D4_i + \beta_5 X1_i + \beta_6 X2_i + \epsilon_i$$

Models with interaction terms between cancer stage and Obesity Level: Model (2):

$$Y_i = \beta_0 + \beta_1 D1_i + \beta_2 D2_i + \beta_3 D3_i + \beta_4 D4_i + \beta_5 X1_i + \beta_6 X2_i + \beta_7 (D1_i * X1_i) + \beta_8 (D2_i * X1_i) + \beta_9 (D3_i * X1_i) + \beta_{10} (D4_i * X1_i) + \epsilon_i$$

Models with interaction terms between cancer stage and genetic risk: Model (3):

$$Y_i = \beta_0 + \beta_1 D1_i + \beta_2 D2_i + \beta_3 D3_i + \beta_4 D4_i + \beta_5 X1_i + \beta_6 X2_i + \beta_7 (D1_i * X2_i) + \beta_8 (D2_i * X2_i) + \beta_9 (D3_i * X2_i) + \beta_{10} (D4_i * X2_i) + \epsilon_i$$

Models with interaction terms between cancer stage, Obesity Level and genetic risk: Model (4):

$$Y_i = \beta_0 + \beta_1 D1_i + \beta_2 D2_i + \beta_3 D3_i + \beta_4 D4_i + \beta_5 X1_i + \beta_6 X2_i + \beta_7 (D1_i * X1_i) + \beta_8 (D2_i * X1_i) + \beta_9 (D3_i * X1_i) + \beta_{10} (D4_i * X1_i) + \beta_{11} (D1_i * X2_i) + \beta_{12} (D2_i * X2_i) + \beta_{13} (D3_i * X2_i) + \beta_{14} (D4_i * X2_i) + \beta_{15} (X1_i * X2_i) + \epsilon_i$$

Where:

- Y_i = Predicted **Target Severity Score**
- β_0 = Intercept (constant term)
- X_{1i} = Obesity Level (numeric)
- X_{2i} = Genetic Risk (numeric)
- $D_{1i} = \begin{cases} 1 & \text{patient is in stage I} \\ 0 & \text{O.W} \end{cases}$
- $D_{2i} = \begin{cases} 1 & \text{patient is in stage II} \\ 0 & \text{O.W} \end{cases}$
- $D_{3i} = \begin{cases} 1 & \text{patient is in stage III} \\ 0 & \text{O.W} \end{cases}$
- $D_{4i} = \begin{cases} 1 & \text{patient is in stage IV} \\ 0 & \text{O.W} \end{cases}$
- ϵ_i = Error term (residuals)

Model (1) summary:

```
Call:
lm(formula = Target_Severity_Score ~ Obesity_Level + Cancer_Stage +
    Genetic_Risk, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.3499 -1.1371  0.0556  1.2037  4.8581

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.288125   0.030174  142.113 < 2e-16 ***
Obesity_Level    0.098638   0.003152   31.290 < 2e-16 ***
Cancer_StageStage I    0.235655   0.028860    8.166 3.31e-16 ***
Cancer_StageStage II   0.488630   0.028796   16.969 < 2e-16 ***
Cancer_StageStage III  0.679658   0.028958   23.471 < 2e-16 ***
Cancer_StageStage IV   0.877912   0.028889   30.389 < 2e-16 ***
Genetic_Risk      0.192566   0.003168   60.792 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.668 on 33498 degrees of freedom
Multiple R-squared:  0.149,    Adjusted R-squared:  0.1489
F-statistic: 977.6 on 6 and 33498 DF,  p-value: < 2.2e-16
```

Model (2) summary:

```
Call:
lm(formula = Target_Severity_Score ~ Obesity_Level * Cancer_Stage +
    Genetic_Risk, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.358 -1.137  0.058   1.204   4.857

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.315241   0.043809  98.501 < 2e-16 ***
Obesity_Level     0.093234   0.007081  13.166 < 2e-16 ***
Cancer_StageStage I    0.193628   0.057363   3.375 0.000738 ***
Cancer_StageStage II   0.449422   0.057516   7.814 5.71e-15 ***
Cancer_StageStage III  0.664879   0.057645  11.534 < 2e-16 ***
Cancer_StageStage IV  0.839920   0.057659  14.567 < 2e-16 ***
Genetic_Risk       0.192518   0.003168  60.765 < 2e-16 ***
Obesity_Level:Cancer_StageStage I  0.008450   0.009967   0.848 0.396528
Obesity_Level:Cancer_StageStage II  0.007867   0.009977   0.788 0.430424
Obesity_Level:Cancer_StageStage III 0.002981   0.010002   0.298 0.765663
Obesity_Level:Cancer_StageStage IV  0.007625   0.010001   0.762 0.445827
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.668 on 33494 degrees of freedom
Multiple R-squared:  0.149,    Adjusted R-squared:  0.1488
F-statistic: 586.6 on 10 and 33494 DF,  p-value: < 2.2e-16
```

Model (3) summary:

```
Call:
lm(formula = Target_Severity_Score ~ Obesity_Level + Cancer_Stage *
    Genetic_Risk, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.3261 -1.1364  0.0575  1.2048  4.8732

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      4.2761454   0.0436852  97.885 < 2e-16 ***
Obesity_Level     0.0986544   0.0031530  31.289 < 2e-16 ***
Cancer_StageStage I    0.2322133   0.0579764   4.005 6.21e-05 ***
Cancer_StageStage II   0.4963284   0.0574090   8.645 < 2e-16 ***
Cancer_StageStage III  0.6861183   0.0575477  11.923 < 2e-16 ***
Cancer_StageStage IV  0.9270984   0.0577450  16.055 < 2e-16 ***
Genetic_Risk       0.1949589   0.0070575  27.624 < 2e-16 ***
Cancer_StageStage I:Genetic_Risk  0.0006244   0.0099969   0.062 0.950
Cancer_StageStage II:Genetic_Risk -0.0015435   0.0100120  -0.154 0.877
Cancer_StageStage III:Genetic_Risk -0.0013034   0.0099781  -0.131 0.896
Cancer_StageStage IV:Genetic_Risk -0.0098549   0.0100295  -0.983 0.326
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.668 on 33494 degrees of freedom
Multiple R-squared:  0.1491,    Adjusted R-squared:  0.1488
F-statistic: 586.7 on 10 and 33494 DF,  p-value: < 2.2e-16
```

Model (4) summary:

```
Call:
lm(formula = Target_Severity_Score ~ Obesity_Level * Cancer_Stage *
    Genetic_Risk, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.3323 -1.1367  0.0546  1.2032  4.8729

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.2630608  0.0810518  52.597  < 2e-16 ***
Obesity_Level 0.1014090  0.0140385   7.224 5.17e-13 ***
Cancer_Stage I 0.2428285  0.1154834   2.103  0.0355 *
Cancer_Stage II 0.5686952  0.1141656   4.981 6.35e-07 ***
Cancer_Stage III 0.8188103  0.1142367   7.168 7.79e-13 ***
Cancer_Stage IV 0.9239841  0.1151957   8.021 1.08e-15 ***
Genetic_Risk  0.2029298  0.0139719  14.524  < 2e-16 ***
Obesity_Level:Cancer_Stage I -0.0022207  0.0200311  -0.111  0.9117
Obesity_Level:Cancer_Stage II -0.0147548  0.0199374  -0.740  0.4593
Obesity_Level:Cancer_Stage III -0.0268372  0.0198090  -1.355  0.1755
Obesity_Level:Cancer_Stage IV  0.0005129  0.0199882   0.026  0.9795
Obesity_Level:Genetic_Risk -0.0016392  0.0024452  -0.670  0.5026
Cancer_Stage I:Genetic_Risk -0.0098123  0.0199114  -0.493  0.6222
Cancer_Stage II:Genetic_Risk -0.0240133  0.0199161  -1.206  0.2279
Cancer_Stage III:Genetic_Risk -0.0309518  0.0197739  -1.565  0.1175
Cancer_Stage IV:Genetic_Risk -0.0168810  0.0199677  -0.845  0.3979
Obesity_Level:Cancer_Stage I:Genetic_Risk  0.0021260  0.0034647   0.614  0.5395
Obesity_Level:Cancer_Stage II:Genetic_Risk  0.0045362  0.0034746   1.306  0.1917
Obesity_Level:Cancer_Stage III:Genetic_Risk  0.0060013  0.0034479   1.741  0.0818 .
Obesity_Level:Cancer_Stage IV:Genetic_Risk  0.0014450  0.0034731   0.416  0.6774
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.668 on 33485 degrees of freedom
Multiple R-squared:  0.1492,    Adjusted R-squared:  0.1487
F-statistic: 309.1 on 19 and 33485 DF,  p-value: < 2.2e-16
```

After fitting the regression models with various interaction terms—between Cancer Stage and Genetic Risk, between Cancer Stage and Obesity Level, and including the three-way interactions—it was observed that the interaction effects were statistically insignificant. The R-squared value remained nearly unchanged across all models, indicating that the inclusion of these interaction terms did not substantially improve the model’s explanatory power. This suggests that the simpler model without interaction terms is more appropriate and parsimonious. Therefore, we can reasonably assume that the effects of Cancer Stage, Obesity Level, and Genetic Risk on the Target Severity Score are additive and independent of each other. This not only supports model interpretability but also reduces the risk of overfitting and multicollinearity associated with unnecessary interaction terms.

But to make sure about this decision, we will apply for Anova test:

```

> #anova model test
> anova(model1, model2)
Analysis of Variance Table

Model 1: Target_Severity_Score ~ Obesity_Level + Cancer_Stage + Genetic_Risk
Model 2: Target_Severity_Score ~ Obesity_Level * Cancer_Stage + Genetic_Risk
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1  33498 93144
2  33494 93140   4    3.0874 0.2776 0.8926
> anova(model1, model3)
Analysis of Variance Table

Model 1: Target_Severity_Score ~ Obesity_Level + Cancer_Stage + Genetic_Risk
Model 2: Target_Severity_Score ~ Obesity_Level + Cancer_Stage * Genetic_Risk
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1  33498 93144
2  33494 93140   4    3.9804 0.3578 0.8387
> anova(model1, model4)
Analysis of Variance Table

Model 1: Target_Severity_Score ~ Obesity_Level + Cancer_Stage + Genetic_Risk
Model 2: Target_Severity_Score ~ Obesity_Level * Cancer_Stage * Genetic_Risk
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1  33498 93144
2  33485 93122 13    21.207 0.5866 0.8671

```

For each ANOVA test comparing the base model (without interaction terms) to a model with interaction:

- Null Hypothesis (H_0): Adding the interaction term does not significantly improve the model (i.e., the interaction coefficient = 0).
- Alternative Hypothesis (H_1): Adding the interaction term significantly improves the model (i.e., the interaction coefficient $\neq 0$).

Anova (model1, model2): Interaction between Cancer Stage and Genetic Risk

- p-value = 0.8926 $\rightarrow$ Not significant ($p > 0.05$)
- Conclusion: No evidence that the interaction between cancer stage and genetic risk improves the model.

Anova (model1, model3): Interaction between Cancer Stage and Obesity Level

- p-value = 0.8387 $\rightarrow$ Not significant ($p > 0.05$)
- Conclusion: No evidence that the interaction between cancer stage and obesity level improves the model.

Anova (model1, model4): Three-way interaction between Cancer Stage, Genetic Risk, and Obesity Level

- p-value = 0.8671 $\rightarrow$ Not significant ($p > 0.05$)
- Conclusion: Even with a three-way interaction, the model does not improve significantly.

Since all p-values are greater than 0.05, we fail to reject the null hypothesis in every case. This suggests that none of the interaction terms significantly contribute to improving the model. Additionally, the R-squared values do not change substantially across models. Therefore, the

model without interaction terms is the most appropriate and parsimonious, and it is reasonable to assume independence between the explanatory variables.

Now, we proceed with answering our research question using the regression model without interaction terms, as it was found to be the most appropriate based on the ANOVA results and model performance metrics. Before drawing conclusions, we will verify that the key assumptions of linear regression linearity, independence of errors, homoscedasticity, normality of residuals, and absence of multicollinearity are satisfied for this model. This will ensure the reliability and validity of our findings.

Initial Model Summary:

```
Call:
lm(formula = Target_Severity_Score ~ Obesity_Level + Cancer_Stage +
    Genetic_Risk, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.3499 -1.1371  0.0556  1.2037  4.8581

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.288125   0.030174  142.113 < 2e-16 ***
obesity_Level    0.098638   0.003152   31.290 < 2e-16 ***
Cancer_StageStage I    0.235655   0.028860    8.166 3.31e-16 ***
Cancer_StageStage II   0.488630   0.028796   16.969 < 2e-16 ***
Cancer_StageStage III  0.679658   0.028958   23.471 < 2e-16 ***
Cancer_StageStage IV  0.877912   0.028889   30.389 < 2e-16 ***
Genetic_Risk     0.192566   0.003168   60.792 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.668 on 33498 degrees of freedom
Multiple R-squared:  0.149,    Adjusted R-squared:  0.1489
```

- All predictors are statistically significant (p-value < 0.001).
- The model's adjusted R^2 is 0.1489, indicating that approximately 14.89% of the variability in severity scores is explained by the predictors.
- The most influential variable is Genetic Risk, followed by Cancer Stage.

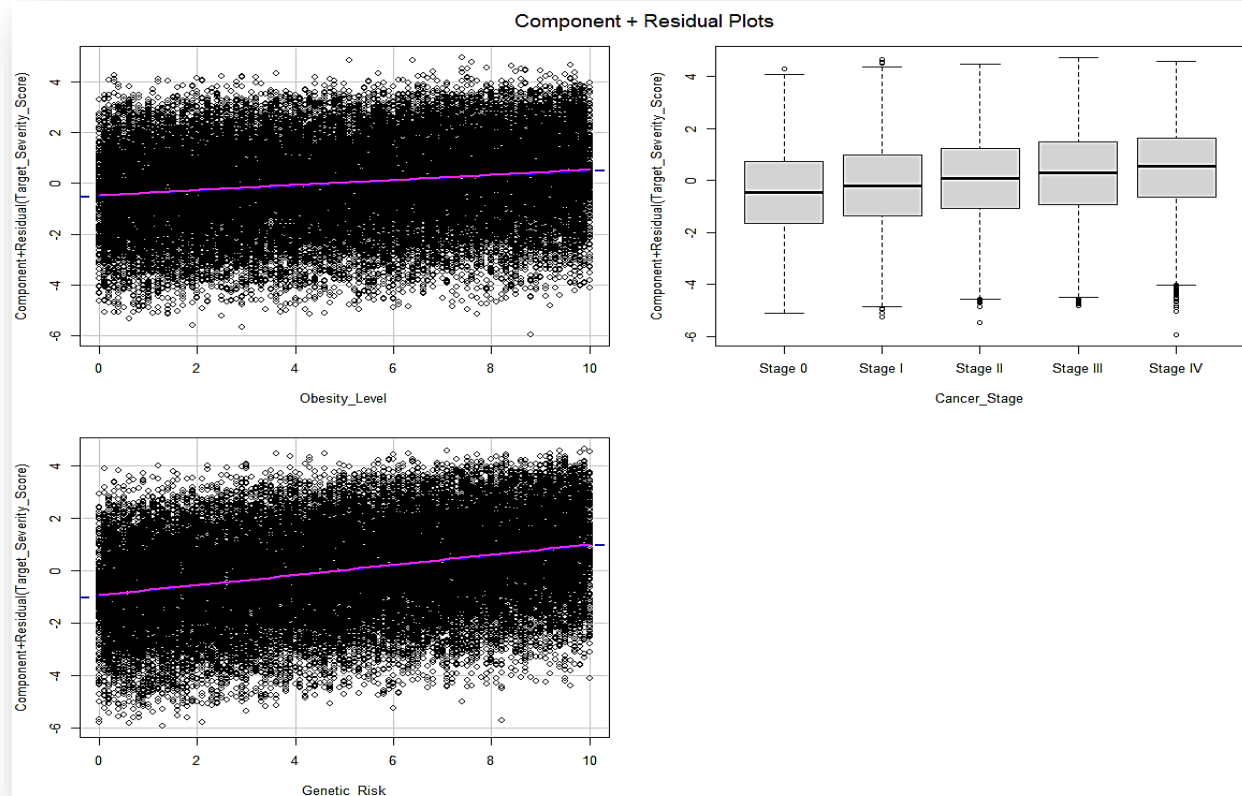
The interpretation of coefficients:

- Interpretation of the coefficients: All coefficients are statistically significant since their p-values are less than 0.05 (in fact, all are < 2e-16). Therefore, we reject the null hypothesis for each coefficient, which states that the coefficient is equal to 0 (i.e., no effect). The interpretations are as follows:
- β_0 (Intercept): The average value of the Target Severity Score for a patient who is in the baseline cancer stage (reference category, e.g., Stage 0), with an Obesity Level and Genetic Risk equal to zero, is approximately 4.2881.

- β_1 (Obesity Level): When the Obesity Level increases by 1 unit, the average Target Severity Score is expected to increase by approximately 0.0986, holding cancer stage and genetic risk constant.
- β_2 (Cancer Stage I): A patient in Stage I is expected to have a Target Severity Score that is, on average, 0.2366 units higher than a patient in the baseline cancer stage, holding Obesity Level and Genetic Risk constant.
- β_3 (Cancer Stage II): A patient in Stage II is expected to have a Target Severity Score that is 0.4868 units higher than a patient in the baseline cancer stage, assuming other variables remain constant.
- β_4 (Cancer Stage III): A patient in Stage III is expected to have a Target Severity Score that is 0.6797 units higher than a patient in the baseline stage, controlling for Obesity Level and Genetic Risk.
- β_5 (Cancer Stage IV): A patient in Stage IV is expected to have a Target Severity Score that is 0.8779 units higher than a patient in the baseline stage, all else held constant.
- β_6 (Genetic Risk): When the Genetic Risk score increases by 1 unit, the average Target Severity Score is expected to increase by approximately 0.1926, assuming Obesity Level and Cancer Stage remain.

• **Assumption Testing:**

1. **Linearity:**



- **Obesity Level Plot:**

The plot assesses the partial relationship between Obesity Level and the response variable after adjusting for other predictors. A roughly linear pattern with residuals evenly scattered around the fitted line indicates that the linearity assumption holds. No systematic curvature or trend suggests the model adequately captures this effect.

- **Cancer Stage Plot:**

As a categorical predictor, the plot displays the adjusted effect of cancer stages on the outcome. Well-separated groups with consistent residual patterns confirm appropriate modeling of this factor. Any non-random pattern within groups might indicate model misspecification or need for alternative coding.

- **Genetic Risk Plot:**

This plot evaluates the linearity of the relationship between Genetic Risk and the dependent variable. A linear trend with residuals symmetrically distributed around the regression line supports the assumption of linearity. Deviations or curvature would imply potential model improvement by variable transformation.

2. Independence of Residuals:

Durbin-Watson test statistic: 2.006 ($p = 0.57$), indicating no autocorrelation problem.

```
> durbinwatsonTest(model)
lag Autocorrelation D-w Statistic p-value
1 -0.003222209 2.0064 0.57
Alternative hypothesis: rho != 0
```

3. Multicollinearity:

Variance Inflation Factors (VIFs) are all below 1.01, suggesting no multicollinearity issues.

```
> vif(model) # VIF values > 5 or 10 may be problematic
          GVIF Df GVIF^(1/(2*Df))
obesity_Level 1.000077 1 1.000039
Cancer_Stage 1.000286 4 1.000036
Genetic_Risk 1.000314 1 1.000157
```


4. Homoscedasticity:

Breusch-Pagan test p-value = 0.003, indicating heteroscedasticity (non-constant variance of residuals).

```
> bptest(model) # p < 0.05 indicates heteroscedasticity
      studentized Breusch-Pagan test
data:  model
BP = 19.748, df = 6, p-value = 0.00307
```

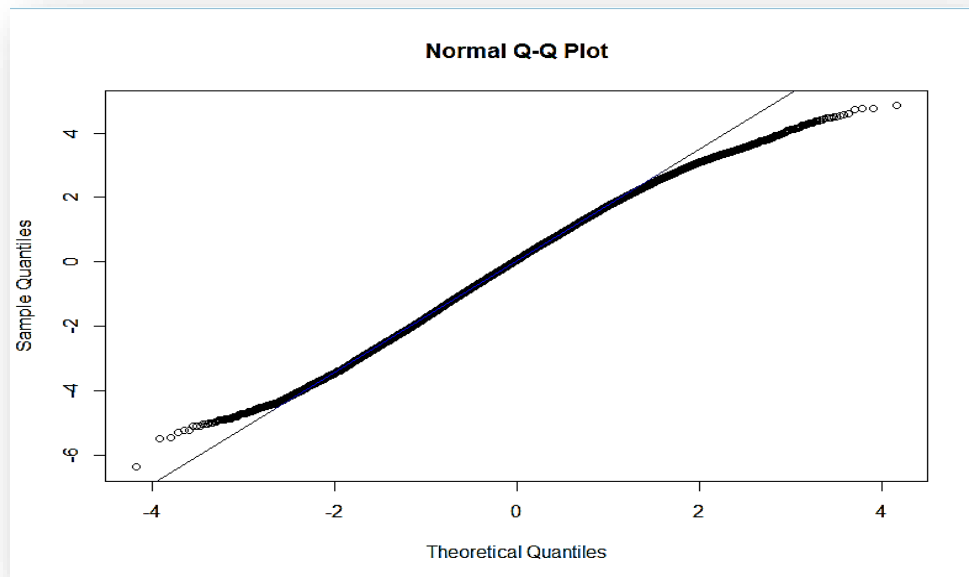
- **Robust Standard Errors Application:**

```
t test of coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.2881252  0.0302464 141.7729 < 2.2e-16 ***
Obesity_Level    0.0986383  0.0031579  31.2349 < 2.2e-16 ***
Cancer_StageStage I  0.2356553  0.0289373   8.1436 3.967e-16 ***
Cancer_StageStage II 0.4886300  0.0289083  16.9028 < 2.2e-16 ***
Cancer_StageStage III 0.6796582  0.0292782  23.2138 < 2.2e-16 ***
Cancer_StageStage IV 0.8779123  0.0287228  30.5650 < 2.2e-16 ***
Genetic_Risk    0.1925664  0.0031696  60.7541 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> bptest(model_clean)
      studentized Breusch-Pagan test
data:  model_clean
BP = 17.197, df = 6, p-value = 0.008585
```

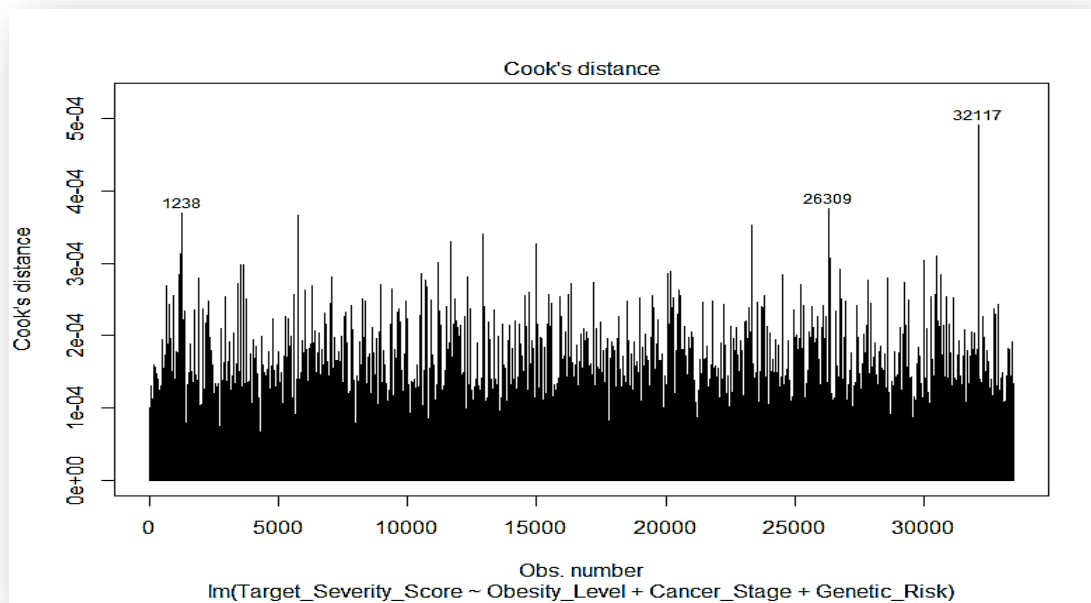
- Used HC3 method to adjust for heteroscedasticity.
- While heteroscedasticity remained present even after outlier removal, robust standard errors (HC3) were employed to ensure valid inference. Thus, the model's conclusions are considered statistically reliable.

5. Normality of Residuals



- Residuals generally follow the reference line, indicating approximate normality in the central region; however, significant deviations at the tails suggest departures from normality. This implies potential issues with extreme values or heavy tails, which might affect inference and require further investigation or robust methods.

- **Outliers Analysis:**



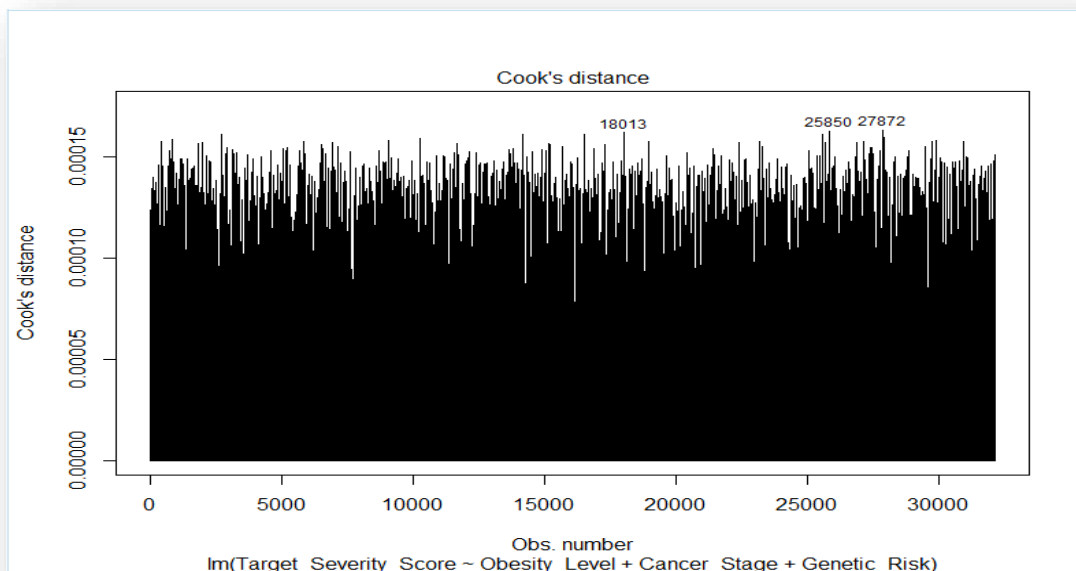
- Observations with Cook's Distance greater than $\frac{4}{n}$ were flagged as outliers.
- These were removed to refine the model.

```
Call:
lm(formula = Target_Severity_Score ~ Obesity_Level + Cancer_Stage +
    Genetic_Risk, data = df_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-3.9933 -1.1014  0.0278  1.1249  3.7875

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.256348   0.028096  151.495  <2e-16 ***
Obesity_Level   0.104719   0.002947   35.532  <2e-16 ***
Cancer_StageStage I   0.260910   0.026806    9.733  <2e-16 ***
Cancer_StageStage II  0.543037   0.026739   20.309  <2e-16 ***
Cancer_StageStage III 0.763378   0.026932   28.345  <2e-16 ***
Cancer_StageStage IV  0.973411   0.026824   36.289  <2e-16 ***
Genetic_Risk    0.194217   0.002960   65.606  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.516 on 32144 degrees of freedom
Multiple R-squared:  0.1852,    Adjusted R-squared:  0.1851
F-statistic: 1218 on 6 and 32144 DF,  p-value: < 2.2e-16
```



After

identifying and excluding influential outliers using Cook's Distance threshold ($<\frac{4}{n}$), the model was refitted on the cleaned dataset.

- R^2 increased from 14.89% to 18.51%, indicating that the refined model explains a larger portion of the variance in the outcome variable.
- Regression coefficients became more stable and interpretable, especially for Cancer Stage, which showed increased statistical significance post-cleaning.

- **Conclusion:**

The multiple linear regression model reveals a clear and statistically significant relationship between cancer severity and the predictors: **obesity level**, **genetic risk**, and **cancer stage**. After identifying and removing influential outliers, the model's explanatory power improved, as reflected by the increase in adjusted R^2 .

Although heteroscedasticity remained present, the application of **robust standard errors (HC3)** ensured valid inference by correcting the standard errors and maintaining the integrity of hypothesis testing. Diagnostic checks further confirmed the model's linearity and absence of multicollinearity.

Overall, the model provides a stable and interpretable framework for understanding the factors associated with cancer severity. It can be considered a reliable predictive tool, especially when complemented with robust statistical techniques to address potential violations of classical regression assumptions.

Q4:"Is there a statistically significant effect of smoking, alcohol use, obesity, Gender on cancer severity scores, Do the associations between these risk factors and severity scores differ by gender? "

Cancer remains one of the leading causes of morbidity and mortality worldwide, with its severity and progression influenced by a complex interplay of genetic, environmental, and lifestyle factors. Among these, modifiable risk factors such as smoking, alcohol consumption, and obesity have been consistently implicated in both the development and progression of various cancers. Studying the association between these risk factors and cancer severity scores is crucial for multiple scientific, clinical, and public health reasons. Furthermore, understanding whether these associations differ by gender adds an essential dimension to personalized medicine and targeted interventions.

Goals of Studying This Research Question:

➤ Understand the Impact of Lifestyle on Cancer Severity

The primary goal of investigating the statistical association between smoking, alcohol use, obesity, and cancer severity scores is to clarify the extent to which these modifiable behaviors influence not only cancer incidence, quantifying the impact of obesity and alcohol use on cancer progression may reveal pathways for lifestyle modifications that can improve survival rates and quality of life.

➤ Identify Gender Differences in Cancer Risk

Another critical goal is to determine whether these associations differ by gender. Biological, hormonal, and behavioral differences between men and women can influence both cancer development and progression, recognizing gender-specific patterns can guide the development of more personalized prevention and treatment approaches.

➤ Improve Patient Care and Treatment Planning

➤ Promote Prevention and Early Intervention

➤ Guide Public Health Policies

➤ Advance Scientific Knowledge

So, I will compare between two models the reduced model which represents the individual effect for each of *smoking habits*, *alcohol use*, *Obesity Level* and *gender* and the full model where I will add the interaction terms that represents the effect of the interaction between them.

The explanatory variables include both quantitative and qualitative variables. Regression models containing mixture of quantitative and qualitative variables are called analysis of covariance (ANCOVA) models

Variables:

Dependent Variable:

- Target_Severity_Score (continuous, higher = worse severity).

Independent Variables:

1. Primary Predictors:

- Smoking (continuous, scaled 0–10).
 - Alcohol_Use (continuous, scaled 0–10).
 - Gender (categorical: Male, Female).
 - Obesity_Level (continuous, scaled 0–10).
- $\text{Gender_Male}_i = \begin{cases} 1 & \text{IF patient is male} \\ 0 & \text{O.W} \end{cases}$

The reduced model (main effect model):

$\text{Target_Severity_Score} = \beta_0 + \beta_1 \text{Smoking} + \beta_2 \text{Alcohol_Use} + \beta_3 \text{Obesity_Level} + \beta_4 \text{Gender_Male} + \epsilon$

- The mean value of Target_Severity_Score for Male cancer patients:

$E(\text{Target_Severity_Score} | \text{Gender_Male}=1) = (\beta_0 + \beta_4) + \beta_1 \text{Smoking} + \beta_2 \text{Alcohol_Use} + \beta_3 \text{Obesity_Level}$

- The mean value of Target_Severity_Score for Female cancer patients:

$E(\text{Target_Severity_Score} | \text{Gender_Male}=0) = \beta_0 + \beta_1 \text{Smoking} + \beta_2 \text{Alcohol_Use} + \beta_3 \text{Obesity_Level}$

R – Output:

```
Call:
lm(formula = Target_Severity_Score ~ Smoking + Alcohol_Use + 
    obesity_Level + Gender, data = cancer_data)

Residuals:
    Min       1Q   Median       3Q      Max 
-4.6117 -1.0223  0.1009  1.0934  3.9218 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.962064   0.026025  113.81  <2e-16 ***
Smoking      0.283134   0.002727  103.81  <2e-16 ***
Alcohol_Use  0.214572   0.002720   78.89  <2e-16 ***
obesity_Level 0.098478   0.002720   36.21  <2e-16 ***
GenderMale   0.521786   0.015720   33.19  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.439 on 33500 degrees of freedom
Multiple R-squared:  0.3665,    Adjusted R-squared:  0.3664 
F-statistic: 4846 on 4 and 33500 DF,  p-value: < 2.2e-16
```

The Estimated model:

$\widehat{\text{Target_Severity_Score}} = 2.962064 + 0.283134(\text{Smoking}) + 0.214572(\text{Alcohol_Use}) + 0.098478(\text{Obesity_Level}) + 0.521786(\text{Gender_Male})$

The Whole Model Significance:

Using F-statistics:

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \quad \text{V.s} \quad H_1: \text{at least one of them} \neq 0$$

Since $F(4, 33500) = 4846$, $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level. The whole model is significant with p-value less than the significance level.

The adjusted R^2 is approximately 34% which means that the “Smoking, Alcohol_Use, Obesity_Level and Gender” can explain 36.6 % of the variation in the “Target_Severity_Score”.

Residuals:

Median (0.1009) is close to zero, and Q1/Q3 are roughly symmetric around zero (-1.02 vs. 1.09). Suggests no major skewness.

The Coefficients Significance:

- $\hat{\beta}_0 = 2.962064$

$$H_0: \beta_0 = 0 \quad \text{V.s} \quad H_1: \beta_0 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_0 is significant.

- $\hat{\beta}_1 = 0.283134$:

$$H_0: \beta_1 = 0 \quad \text{V.s} \quad H_1: \beta_1 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_1 is significant.

- $\hat{\beta}_2 = 0.214572$:

$$H_0: \beta_2 = 0 \quad \text{V.s} \quad H_1: \beta_2 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_2 is significant.

- $\hat{\beta}_3 = 0.098478$:

$$H_0: \beta_3 = 0 \quad \text{V.s} \quad H_1: \beta_3 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_3 is significant.

- $\hat{\beta}_4 = 0.521786$:

$$H_0: \beta_4 = 0 \quad \text{V.s} \quad H_1: \beta_4 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_4 is significant.

Coefficients Interpretations:

- $\hat{\beta}_0 = 2.962064$:

The mean value of Target_Severity_Score for female (reference category) cancer patient is about 2.962064, when all predictors (Smoking, Alcohol_Use, Obesity level) are zero (statistically significant).

- $\hat{\beta}_1 = 0.283134$:

The average change in Target_Severity_Score value is 0.282617 when Smoking increases by 1 unit, holding Alcohol_Use, Gender constants (statistically significant).

- $\hat{\beta}_2 = 0.214572$:

The average change in Target_Severity_Score value is 0.214572 when Alcohol_Use increases by 1 unit, holding Smoking, Obesity Level and Gender constants. (statistically significant).

- $\hat{\beta}_3 = 0.098478$:

The average change in Target_Severity_Score value is 0.098478 when Obesity_Level increases by 1 unit, holding Smoking, Alcohol_Use and Gender constants. (statistically significant).

- Differential Intercept ($\hat{\beta}_4 = 0.521786$):

The average value of Target_Severity_Score for Male cancer patients is higher than that for Female cancer patients by 0.214572 holding the Smoking, Alcohol_Use and Obesity_Level constants (Statistically significant).

The full model (interaction model):

$\text{Target_Severity_Score} = \beta_0 + \beta_1 \text{Smoking} + \beta_2 \text{Alcohol_Use} + \beta_3 \text{Obesity_Level} + \beta_4 \text{Gender_Male} + \beta_5 (\text{Smoking} * \text{Gender_Male}) + \beta_6 (\text{Alcohol_Use} * \text{Gender_Male}) + \beta_7 (\text{Obesity_Level} * \text{Gender}) + \epsilon$.

- The mean value of Target_Severity_Score for Male cancer patients:

$E(\text{Target_Severity_Score} | \text{Gender_Male}=1) = (\beta_0 + \beta_4) + (\beta_1 + \beta_5) \text{Smoking} + (\beta_2 + \beta_6) \text{Alcohol_Use} + (\beta_3 + \beta_7) \text{Obesity_Level}$

- The mean value of Target_Severity_Score for Female cancer patients:

$E(\text{Target_Severity_Score} | \text{Gender_Male}=0) = \beta_0 + \beta_1 \text{Smoking} + \beta_2 \text{Alcohol_Use} + \beta_3 \text{Obesity_Level}$

R Output:

```
Call:
lm(formula = Target_Severity_Score ~ Smoking * Gender + Alcohol_Use *
    Gender + Obesity_Level * Gender, data = cancer_data)

Residuals:
    Min       1Q   Median       3Q      Max
-4.5932 -1.0201  0.1018  1.0923  3.9027

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.916956   0.035130   83.033  <2e-16 ***
Smoking         0.284266   0.003878   73.304  <2e-16 ***
GenderMale      0.611789   0.049565   12.343  <2e-16 ***
Alcohol_Use     0.219312   0.003861   56.809  <2e-16 ***
Obesity_Level   0.101635   0.003834   26.507  <2e-16 ***
Smoking:GenderMale -0.002250  0.005455   -0.412  0.6801
GenderMale:Alcohol_Use -0.009414  0.005440   -1.731  0.0835
GenderMale:Obesity_Level -0.006357  0.005440   -1.169  0.2426
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.439 on 33497 degrees of freedom
Multiple R-squared:  0.3666,    Adjusted R-squared:  0.3665
F-statistic: 2770 on 7 and 33497 DF,  p-value: < 2.2e-16
```

The Estimated Model:

$\widehat{\text{Target_Severity_Score}} = 2.91656 + 0.284266 \text{ Smoking} + 0.219312 \text{ Alcohol_Use} + 0.101635 \text{ Obesity_Level} + 0.611789 \text{ Gender_Male} - 0.002250 (\text{Smoking} * \text{Gender_Male}) - 0.009414 (\text{Alcohol_Use} * \text{Gender_Male}) - 0.006357 (\text{Obesity_Level} * \text{Gender})$.

The Whole Model Significance:

Using F-statistics:

$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0$

H_1 : at least one of them $\neq 0$

Since $F(7, 33497) = 2770$, $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level. The whole model is significant with $p\text{-value}$ less than the significance level.

The adjusted R^2 is approximately 36.6 % which means that the “Smoking, Alcohol_Use and Obesity_Level and Gender” can explain 36.6 % of the variation in the “Target_Severity_Score” (virtually unchanged from non-interaction model).

Residuals: Symmetric around zero (median = 0.1018).

The Coefficients Significance:

- $\widehat{\beta_0} = 2.91656$

$H_0: \beta_0 = 0$ V.s $H_1: \beta_0 \neq 0$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_0 is significant.

- $\widehat{\beta_1} = 0.284266$:

$$H_0: \beta_1 = 0 \quad \text{V.s} \quad H_1: \beta_1 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_1 is significant.

- $\widehat{\beta_2} = 0.219312$:

$$H_0: \beta_2 = 0 \quad \text{V.s} \quad H_1: \beta_2 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_2 is significant.

- $\widehat{\beta_3} = 0.101635$:

$$H_0: \beta_3 = 0 \quad \text{V.s} \quad H_1: \beta_3 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} < 0.01$, the null hypothesis is rejected at 1% significance level, β_3 is significant.

- $\widehat{\beta_4} = 0.611789$:

$$H_0: \beta_4 = 0 \quad \text{V.s} \quad H_1: \beta_4 \neq 0$$

Since $p\text{-value} = 2.2e^{-16} > 0.01$, the null hypothesis is rejected at 1% significance level, β_4 is significant.

- $\widehat{\beta_5} = -0.002250$:

$$H_0: \beta_5 = 0 \quad \text{V.s} \quad H_1: \beta_5 \neq 0$$

Since $p\text{-value} = 0.6801 > 0.01$, the null hypothesis is not rejected at 1% significance level, β_5 is insignificant.

- $\widehat{\beta_6} = -0.009414$:

$$H_0: \beta_6 = 0 \quad \text{V.s} \quad H_1: \beta_6 \neq 0$$

Since $p\text{-value} = 0.0835 > 0.01$, the null hypothesis is not rejected at 1% significance level, β_6 is insignificant.

- $\widehat{\beta_7} = -0.006357$:

- $H_0: \beta_7 = 0 \quad \text{V.s} \quad H_1: \beta_7 \neq 0$

Since $p\text{-value} = 0.2426 > 0.01$, the null hypothesis is not rejected at 1% significance level, β_7 is insignificant.

Coefficients Interpretations:

- Intercept ($\widehat{\beta_0} = 2.91656$):

The mean value of Target_Severity_Score for female (reference category) cancer patient is about 2.91656 when all predictors (Smoking, Alcohol_Use, Obesity_level) are zero (statistically significant).

- $\hat{\beta}_1 = 0.284266$:

The average change in Target_Severity_Score value is 0.284266 for Female cancer patients when Smoking increases by 1 unit holding Alcohol_Use and Obesity_Level constants (statistically significant).

- $\hat{\beta}_2 = 0.219312$:

The average change in Target_Severity_Score value is 0.219312 for Female cancer patients when Alcohol_Use increases by 1 unit holding Smoking and Obesity_Level constants, (statistically significant).

- $\hat{\beta}_3 = 0.101635$:

The average change in Target_Severity_Score value is 0.101635 for Female cancer patients when Obesity_Level increases by 1 unit holding Smoking and Alcohol_Use constants, (statistically significant).

- Differential Intercept ($\hat{\beta}_4 = 0.611789$):

The difference between the average value of Target_Severity_Score for Male cancer patients and Female cancer patients is 0.611789 when all predictors (Smoking, Alcohol_Use and Obesity_Level) are zero (Statistically significant).

Interaction Terms:

- Differential Slope ($\hat{\beta}_5 = -0.002250$):

The average change in Target_Severity_Score of Male cancer patients is lower than that of Female cancer patients by 0.00225 when smoking increases by 1 unit, holding Alcohol_Use and Obesity_Level constants (statistically insignificant).

- Differential Slope ($\hat{\beta}_6 = -0.009414$)

The average change in Target_Severity_Score of Male cancer patients is lower than that of Female cancer patients by 0.009414 when Alcohol_Use increases by 1 unit, holding Smoking and Obesity_Level constants (statistically insignificant).

- Differential Slope ($\hat{\beta}_7 = -0.006357$):

The average change in Target_Severity_Score of Male cancer patients is lower than that of Female cancer patients by 0.006357 when Obesity_Level increases by 1 unit, holding Smoking and Alcohol_Use constants (statistically insignificant)

- No significant interactions: None of the interaction terms are statistically significant (all $p > 0.01$). The impact of smoking, alcohol, and obesity on severity is similar for males and females.

Choosing The best Model:

Using the F-Test for Nested Models and Model Selection

The F-test for nested models compares a simpler model (reduced) to a more complex one (full) to determine if the additional terms significantly improve the fit.

$$H_0: \beta_5 = \beta_6 = \beta_7 = 0$$

$$H_1: \text{at least one of them} \neq 0$$

$$F = \frac{(ESS_{Full} - ESS_{reduced})/3}{RSS(Full)/(n - 8)}$$

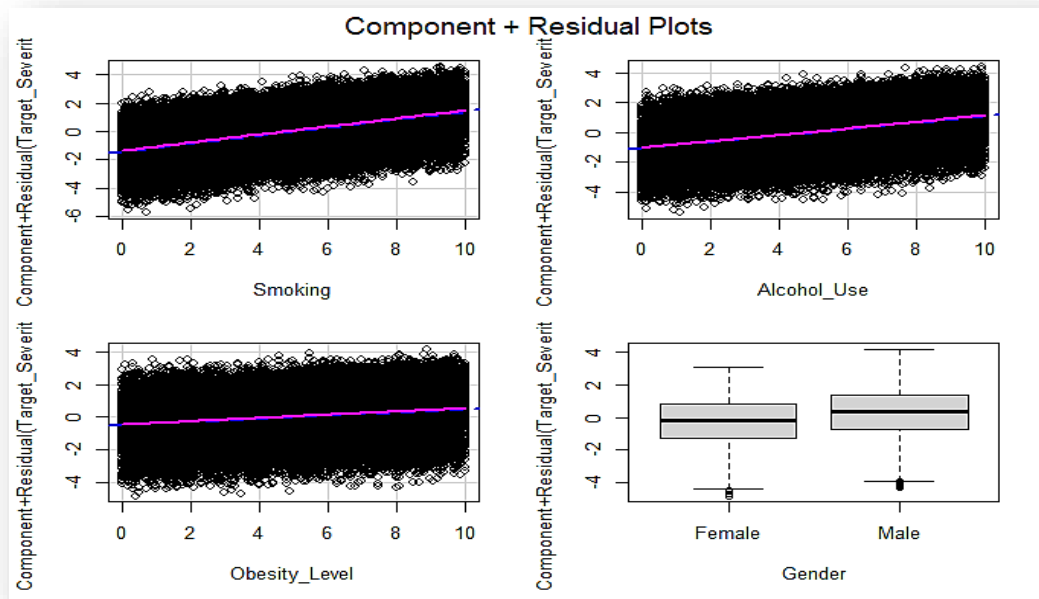
- Since (F-statistic = 1.5161, with a p-value = 0.2081 > 0.01), the null hypothesis is not rejected at 1% significance level, indicating that the interaction terms do not significantly improve the model fit. (The reduced model (main effects only) is preferable as it is simpler and equally effective at predicting severity score).

```
> anova(reduced_model,full_model )
Analysis of Variance Table

Model 1: Target_Severity_Score ~ Smoking + Alcohol_Use + Obesity_Level +
  Gender
Model 2: Target_Severity_Score ~ Smoking * Gender + Alcohol_Use * Gender +
  Obesity_Level * Gender
  Res.Df  RSS Df Sum of Sq    F Pr(>F)
1  33500 69336
2  33497 69327   3    9.4132 1.5161 0.2081
```

Checking the assumptions:

i. Linearity:



➤ Smoking vs Target_Severity_Score:

The magenta smoothed line shows a clear positive linear trend, suggesting a reasonably linear relationship. The linearity assumption appears valid for Smoking.

➤ Alcohol_Use vs Target_Severity_Score:

The smoothed line again shows a moderate positive linear trend with consistent spread across the x-axis. The linearity assumption is reasonably valid for Alcohol_Use.

➤ Obesity_Level vs Target_Severity_Score:

The smoothed line is almost flat, suggesting a weak or no clear linear trend. The linearity assumption is less supported here, implying that Obesity_Level may have a weak or nonlinear effect on the response variable.

➤ Gender vs Target_Severity_Score (Bottom Right):

This is a boxplot (since Gender is categorical). There's a slight difference in the medians and spread between Female and Male. Linearity does not apply to categorical variables in the same way. Instead, this shows that Gender may have a marginal effect, but its inclusion as a factor is appropriate, Gender is categorical and doesn't violate linearity assumptions.

ii. Independence of errors:

Using Durbin-Watson Test

Null Hypothesis (H_0):

There is no first-order autocorrelation in the residuals.

Alternative Hypothesis (H_1):

There is first-order autocorrelation in the residuals.

```
Durbin-watson test  
data: reduced_model  
Dw = 2.0134, p-value = 0.8895  
alternative hypothesis: true autocorrelation is greater than 0
```

Since $p\text{-value} = 0.8895 > 0.01$, then the null hypothesis is not rejected, there is no significant first-order autocorrelation in the residuals. This means the residuals are independent.

This is a desirable outcome for a regression model, as it indicates that the assumption of independent errors is likely met. This helps ensure that the standard errors of the regression coefficients are unbiased and that the p-values for individual predictors are reliable.

iii. Multicollinearity:

```
> vif(reduced_model)  
      smoking      Alcohol_Use      obesity_Level      Gender  
1.000055      1.000054      1.000069      1.000040
```

Since all VIFs ≈ 1 , indicating no multicollinearity issues among the predictors. The model's predictors (**Smoking**, **Alcohol_Use**, **Obesity_level**, **Gender**) are statistically independent of one another in their explanatory power, this is ideal for regression analysis, as multicollinearity can distort coefficient estimates and p-values.

iv. Homoscedasticity:

Using Studentized Breusch-Pagan Test for Heteroscedasticity

```
studentized Breusch-Pagan test  
data: reduced_model  
BP = 47.582, df = 4, p-value = 1.153e-09
```

Null Hypothesis (H_0): Homoscedasticity (residuals have constant variance)

Alternative Hypothesis (H_1): Heteroscedasticity (residual variance depends on predictors or fitted values)

Since $p_value = 1.153e^{-9} < 0.01$, the null hypothesis is rejected at 1 % significance level, Strong evidence of heteroscedasticity in the model. Residual variance changes systematically with predictors/fitted values.

Address Heteroscedasticity:

Using robust standard errors:

The use of HC standard errors ensures that p-values and confidence intervals remain reliable despite heteroscedasticity.

```
t test of coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.9620642  0.0256517 115.472 < 2.2e-16 ***
Smoking      0.2831343  0.0027235 103.958 < 2.2e-16 ***
Alcohol_Use  0.2145723  0.0027111  79.145 < 2.2e-16 ***
Obesity_Level 0.0984778  0.0027230  36.165 < 2.2e-16 ***
GenderMale   0.5217856  0.0157176  33.197 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We could solve this problem by using the following code:

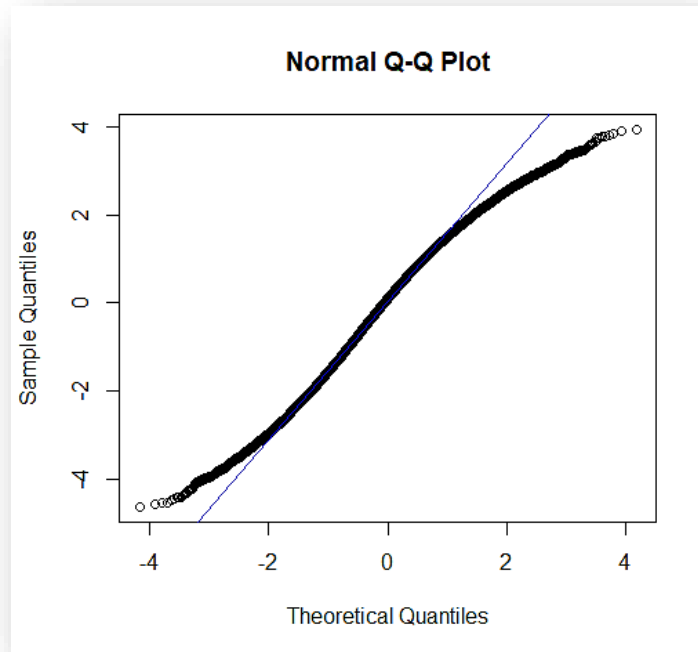
```
#Robust standard errors
library(sandwich)
library(lmtest)
coefTest(reduced_model, vcov = vcovHC(reduced_model, type = "HC1")) # Huber-white SES
install.packages("estimatr")
#Load the package
library(estimatr)
# Fit model with robust SES
reduced_model_robust <- lm_robust(Target_Severity_Score ~ Smoking + Alcohol_Use + Obesity_Level + Gender , data = cancer_data)
# Summary
summary(reduced_model_robust)
plot(fitted(reduced_model), residuals(reduced_model))
abline(h = 0, col = "red")
```

v. Normality:

The Q-Q (Quantile-Quantile) plot compares the distribution of the model's residuals to a theoretical normal distribution.

The Normal Q-Q plot reveals that the residuals of the model deviate from a perfect normal distribution, particularly in the tails. While the central portion of the data aligns reasonably well with the theoretical normal quantiles, the left and right tails show significant departures:

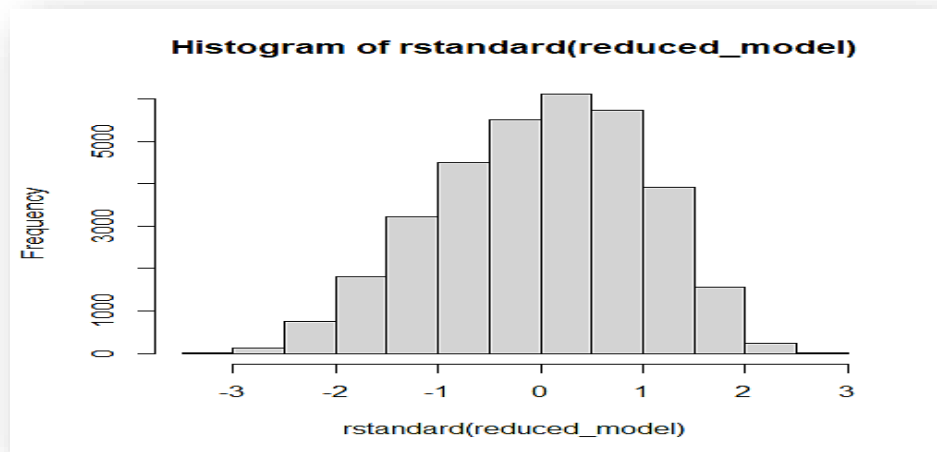
- Left tail: Residuals are more extreme (negative) than expected, suggesting left-skewness or heavy-tailed behavior.
- Right tail: Residuals are less extreme (positive) than expected, indicating potential right-skewness or light-tailed behavior.



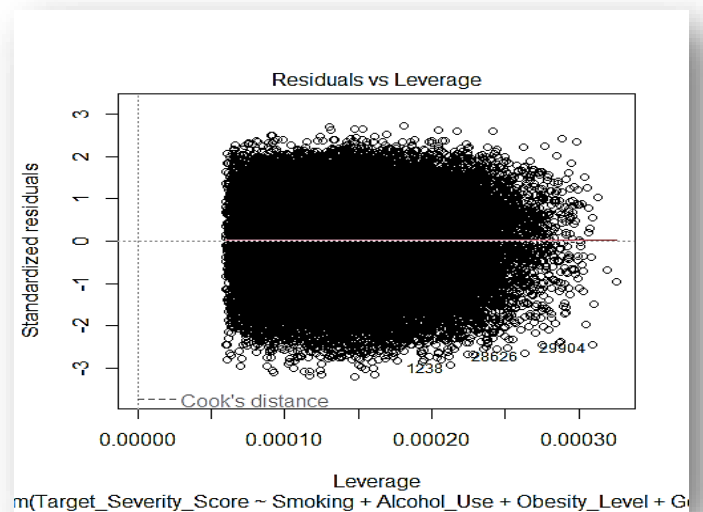
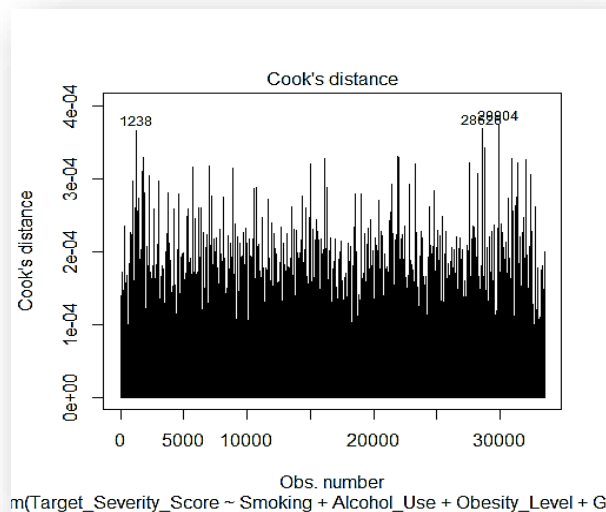
The Q-Q plot clearly indicates that the residuals violate the normality assumption, with significant deviations in both tails. Despite attempts to address this through transformations or robust methods, the underlying non-normality persists, likely due to the inherent distribution of the data or unmodeled complexities. Given the large sample size ($N=33,500$), the Central Limit Theorem ensures that regression coefficient estimates and p-values remain asymptotically valid, making the model's inferences reasonably reliable despite the non-normality. However, predictions for extreme values may be less accurate, and confidence intervals could be slightly mis calibrated in the tails. Since the non-normality appears systematic and unfixable through standard approaches. The analysis should proceed with this limitation acknowledged, emphasizing robust standard errors and focusing on the model's primary inferences rather than precise tail behavior.

Histogram of The Residuals:

The histogram of standardized residuals from the reduced model clearly demonstrates a right-skewed distribution, confirming the non-normality previously observed in the Q-Q plot



vi. Influential observations:



- All Cook's distance values are extremely low (≤ 0.0004), far below the typical warning threshold of 0.5 (conservative) or 1 (strict). No observations exert significant influence on the regression coefficients. The Cook's distance and Residuals vs Leverage plots provide valuable insights into the influence of individual data points on the regression model. The Cook's distance values are all extremely low, falling well below the standard thresholds of 0.5 or 1, indicating that no single observation has a disproportionate impact on the model's coefficients. This suggests the model is robust and the estimated coefficients reliably represent the overall dataset. The absence of highly influential points means the regression results are not being driven by outliers or extreme values, reinforcing the stability of the model.

Conclusion:

This study investigated the effects of smoking, alcohol use, obesity, and gender on cancer severity scores, and whether these relationships differ by gender. The results showed that all four factors significantly predicted cancer severity, with smoking having the strongest effect. However, no significant gender interactions were found, meaning these risk factors affect both men and women similarly. The reduced model without interaction terms was selected as it performed equally well while being simpler.

The analysis confirmed that the model met key statistical assumptions, with no issues of multicollinearity or influential outliers. While minor deviations from normality and homoscedasticity were detected, the large sample size ensured reliable results. These findings suggest that smoking and alcohol use are important predictors of cancer severity across genders, whereas obesity plays a smaller role.

These results highlight the need for gender-neutral interventions targeting smoking and alcohol use to reduce cancer severity. The consistent gender difference in severity scores calls for further research into underlying causes. Overall, this study provides valuable insights into how modifiable lifestyle factors influence cancer outcomes, supporting evidence-based prevention strategies.

- **Final Conclusion:**

This study examined how cancer severity varies depending on the patient's gender and cancer stage, while also considering age and exposure to air pollution. The results showed that patients in advanced stages of cancer suffer from more severe cases, which aligns with expectations. It was also found that males have higher average severity scores than females. In addition, older individuals and those living in more polluted areas tend to suffer from more severe cancer cases. These results highlight the importance of early diagnosis and show that environmental factors like air quality should be a public health priority. Although these four factors are important, there are certainly other factors that affect cancer severity and were not included in this analysis.

In the second part of the study, we focused on three factors often outside the patient's control: genetic risk, air pollution, and treatment costs. Each of these factors was found to be significantly associated with cancer severity. The results showed that individuals with a higher genetic predisposition, those exposed to more pollution, or those who faced higher financial burdens for treatment were more likely to experience more severe cases. These findings suggest that health is not only related to personal choices, but also to a person's environment, genetics, and living conditions. This confirms the need for stronger environmental protection, more equitable healthcare systems, and special support for individuals with a family history of the disease.

The third analysis addressed the relationship between obesity, genetic risk, and cancer stage on one side, and cancer severity on the other. The results showed that each of the three factors had a significant impact on severity. Higher obesity or genetic risk levels were associated with worse outcomes, and the disease became more severe at each advanced stage. We also tested whether these factors interact with each other, such as whether the effect of obesity varies according to cancer stage, but we did not find strong evidence of significant interaction. After removing some extreme values that affected the results, the models became more accurate and clearer. These results confirm that each of the three factors independently plays a role in explaining cancer severity, which strengthens the importance of using these indicators to improve early detection, guide treatment plans, and develop public health policies.

Finally, a part of the analysis examined how lifestyle factors and gender affect cancer severity. The focus was on smoking, alcohol consumption, obesity, and the patient's gender. The results showed that all four of these factors are associated with increased severity. Patients who smoke, consume alcohol, or have high obesity levels were more likely to experience more severe cases. It was also found that males suffer from more severe cases than females on average. However, the study showed that the effects of smoking, alcohol, and obesity on cancer severity do not differ much between males and females, meaning these factors affect patients similarly regardless of gender. These results confirm the importance of addressing lifestyle changes to prevent more severe cancer cases, regardless of whether the patient is male or female.

- **Dataset reference:**

<https://www.kaggle.com/datasets/zahidmughal2343/global-cancer-patients-2015-2024>

- Appendix:

1. introduction R code:

```

1 # Install (if not installed) and load necessary packages
2 install.packages("summarytools") # For descriptive summaries
3 install.packages("ggplot2")      # For visualizations
4 install.packages("dplyr")        # For data manipulation
5 install.packages("readxl")       # For reading Excel files
6 install.packages("corrplot")
7 # Load libraries
8 library(summarytools)
9 library(ggplot2)
10 library(dplyr)
11 library(readxl)
12 library(skimm)
13 library(corrplot)
14 # Load your Excel file
15 data <- read_excel("C:/Users/McIt/Downloads/cancer_modified (1).xlsx")
16 # View structure of the dataset
17 str(data)
18 # View the first few rows
19 head(data)
20 # Summary of dataset
21 summary(data)
22 # Select only numeric columns
23 num_data <- select_if(data, is.numeric)
24 # Summary stats: mean, SD, min, max
25 summary(num_data)
26 # Frequency of Cancer Stage
27 table(data$Cancer_Stage)
28 # Frequency of Cancer Type
29 table(data$Cancer_Type)
30 # Frequency of gender
31 table(data$Gender)
32 # Frequency of Country Region
33 table(data$Country_Region)
34 # graphs of the data
35 # 1 Distribution of Target Severity Score
36 # histogram of dependent variable
37 ggplot(data, aes(x = Target_Severity_Score)) +
38   geom_histogram(fill = "skyblue", color = "black", bins = 30) +
39   labs(title = "Distribution of Target Severity Score", x = "Target Severity Score", y = "Count")
40 #2 Correlation Matrix Analysis
41 # Select relevant numeric variables
42 numeric_vars <- data %>%
43   select(Age, Treatment_Cost_USD, Survival_Years, Target_Severity_Score,
44          Genetic_Risk, Air_Pollution, Obesity_Level, Alcohol_Use)
45 # Calculate correlation matrix
46 cor_matrix <- cor(numeric_vars, use = "complete.obs")
47 # Plot heatmap
48 corrplot(cor_matrix, method = "color", type = "upper",
49          tl.col = "black", tl.srt = 45, addCoef.col = "black",
50          col = colorRampPalette(c("blue", "white", "red"))(200))
51 # 3 Temporal Distribution of Patients (2015-2024):
52 # Count patients per year
53 yearly_counts <- data %>%
54   group_by(Year) %>%
55   summarise(Patient_Count = n())
56 # Line plot with labels
57 ggplot(yearly_counts, aes(x = Year, y = Patient_Count)) +
58   geom_line(color = "blue", size = 1.2) +
59   geom_point(color = "red", size = 3) +
60   geom_text(aes(label = Patient_Count), vjust = -0.5, size = 4) + # Add value labels
61   scale_x_continuous(breaks = seq(min(yearly_counts$Year), max(yearly_counts$Year), by = 1)) +
62   labs(title = "Number of Patients by Year",
63        x = "Year",
64        y = "Patient Count") +
65   theme_minimal()
66 # 4 Number of Patients by Age Group:
67 # Add age group column
68 data <- data %>%
69   mutate(Age_Group = cut(Age,
70                          breaks = c(-Inf, 30, 50, 70, Inf),
71                          labels = c("Young (<=30)", "Middle Age (31-50)", "Senior (51-70)", "Elderly (71+)"),
72                          right = TRUE))
73 # Count patients per age group
74 age_group_counts <- data %>%
75   group_by(Age_Group) %>%
76   summarise(Patient_Count = n())
77 # Line plot (x is categorical here, so treat it like factor)
78 ggplot(age_group_counts, aes(x = Age_Group, y = Patient_Count, group = 1)) +
79   geom_line(color = "purple", size = 1.2) +
80   geom_point(color = "darkblue", size = 3) +
81   geom_text(aes(label = Patient_Count), vjust = -0.5, size = 4) +
82   labs(title = "Number of Patients by Age Group",
83        x = "Age Group",
84        y = "Patient Count") +
85   theme_minimal()
86 ###Cleaning the data:
87 # Count of missing values per column
88 colSums(is.na(data))
89 # Function to detect outliers using IQR

```

```

89 # Function to detect outliers using IQR
90 detect_outliers <- function(x) {
91   Q1 <- quantile(x, 0.25, na.rm = TRUE)
92   Q3 <- quantile(x, 0.75, na.rm = TRUE)
93   IQR <- Q3 - Q1
94   return(x < (Q1 - 1.5 * IQR) | x > (Q3 + 1.5 * IQR))
95 }
96 # Apply to numeric variables
97 num_data <- select_if(data, is.numeric)
98 # Count how many outliers in each column
99 outlier_counts <- sapply(num_data, function(x) sum(detect_outliers(x), na.rm = TRUE))
100 outlier_counts
101 # Remove outliers in Target_Severity_Score
102 data_no_outliers <- data %>%
103   filter(!detect_outliers(Target_Severity_Score))
104 }

```

2. Question 1 R code:

```

1  install.packages("car")
2  install.packages("emmeans")
3  install.packages("tidyverse")
4  library(car) # For ANOVA (Type III SS)
5  library(emmeans) # For post-hoc comparisons
6  library(tidyverse) # For data manipulation
7  #####
8  # Load your dataset (replace with your file path)
9  library(readxl)
10 data <- read_excel("cancer.xlsx")
11 View(data)
12 # Check structure (ensure correct variable types)
13 data$Cancer_Stage_Grouped <- ifelse(data$Cancer_Stage == "Stage 0", "Stage_0",
14   ifelse(data$Cancer_Stage %in% c("Stage I", "Stage II"), "Stage_1_2",
15     ifelse(data$Cancer_Stage %in% c("Stage III", "Stage IV"), "Stage_3_4", NA)))
16 data$Age <- as.numeric(data$Age)
17 data$Air_Pollution <- as.numeric(data$Air_Pollution)
18 #####
19 table(data$Cancer_Stage)
20 # Fit ANCOVA model
21 model1 <- lm(Target_Severity_Score ~ Cancer_Stage_Grouped + Gender + Age + Air_Pollution, data = data)
22 summary(model1)
23 model2 <- lm(Target_Severity_Score ~ Cancer_Stage_Grouped * Gender + Age + Air_Pollution, data = data)
24 summary(model2)
25 model3 <- lm(Target_Severity_Score ~ Cancer_Stage_Grouped + Gender * Age + Air_Pollution, data = data)
26 summary(model3)
27 model4 <- lm(Target_Severity_Score ~ Cancer_Stage_Grouped + Gender * Air_Pollution + Age, data = data)
28 summary(model4)
29 model_full <- lm(Target_Severity_Score ~ Cancer_Stage_Grouped * Gender * Age * Air_Pollution, data = data)
30 summary(model_full)
31 ##to compare between the interaction models
32 anova(model2, model3, model4)
33 AIC(model2, model3, model4)
34 ##to choose the best model
35 anova(model1, model4)
36 AIC(model1, model4)
37 #Assumption checks
38 install.packages("lme4")
39 install.packages("lmerTest")
40 library(car)
41 library(lme4)
42 # 1. Linearity
43 crPlots(model1)
44 # 2. Independence of errors
45 durbinWatsonTest(model1)
46 # 3. Multicollinearity
47 vif(model1)
48 # 4. Homoscedasticity
49 bptest(model1)
50 #Robust standard errors (HC3 is a common choice)
51 install.packages("sandwich")
52 library(sandwich)
53 library(lme4)
54 coefTest(model1, vcov = vcovHC(model4, type = "HC3"))
55 # Install if needed
56 install.packages("estimatr")
57 # Load the package
58 library(estimatr)
59 # Fit model with robust SEs
60 model1_robust <- lm_robust(Target_Severity_Score ~ Cancer_Stage_Grouped + Gender + Age + Air_Pollution, data = data)
61 # Summary
62 summary(model1_robust)
63 plot(fitted(model1), residuals(model1))
64 abline(h = 0, col = "red")
65 # 5. Normality (using QQ plot)
66 qqnorm(residuals(model1))
67 qqline(residuals(model1), col = "blue")
68 # 6. Influential observations
69 plot(model1, which = 4)
70 nrow(data)
71 table(data$Gender)
72 table(data$Cancer_Stage_Grouped)
73 table(data$Age)
74 table(data$Air_Pollution)
75

```

3. Question 2 R code:

```
1 # download libraries:
2 library(readxl)
3 library(dplyr)
4 library(ggplot2)
5 #upload data:
6 library(readxl)
7 data <- read_excel("C:/Users/D.Ahmed/Desktop/datasets/cancer_modified.xlsx")
8 # type of data:
9 data <- data %>%
10   mutate(
11     Gender = as.factor(Gender),
12     Age = as.numeric(Age),
13     Survival_Years = as.numeric(Survival_Years)
14   )
15 # 1. model without interaction:
16 model_no_interaction <- lm(Target_Severity_Score ~ Genetic_Risk+Air_Pollution+log(Treatment_Cost_USD), data = data)
17 summary(model_no_interaction)
18 # Assumption checks
19 library(car)
20 library(lmtest)
21 # 1. Residual plot
22 plot(fitted(model_no_interaction), residuals(model_no_interaction))
23 abline(h = 0, col = "red")
24 # 2. Durbin-Watson test
25 library(lmtest)
26 dwtest(model_no_interaction)
27 # 3. Breusch-Pagan test
28 bptest(model_no_interaction)
29 # 4. Normality of residuals
30 qqnorm(residuals(model_no_interaction))
31 qqline(residuals(model_no_interaction))
32 shapiro.test(residuals(model_no_interaction))
33 # 5. VIF for multicollinearity
34 library(car)
35 vif(model_no_interaction)
36 # Run Breusch-Pagan test for heteroskedasticity
37 bp_result <- bptest(model)
38 print(bp_result)
```

4. Question 3 R code:

```
1 library(readxl)
2 library(dplyr)
3 library(ggplot2)
4 library(car)
5 library(lmtest)
6 library(sandwich)
7
8 #Step 1: Load and Prepare the Data
9
10 df <- read_excel("C:/Users/10/Downloads/cancer_modified (1).xlsx")
11 #Convert Cancer_Stage to factor
12 df$Cancer_Stage <- as.factor(df$Cancer_Stage)
13 #Remove missing values
14 df <- na.omit(df)
15
16 # Step 2: Fit the Base Model (No Interaction)
17
18 model_base <- lm(Target_Severity_Score ~ Obesity_Level + Cancer_Stage + Genetic_Risk, data = df)
19 summary(model_base)
20
21 # Step 3: Fit Models with Interaction Terms
22
23 # Model 1: Cancer_Stage ? Genetic_Risk
24 model_interact1 <- lm(Target_Severity_Score ~ Obesity_Level + Cancer_Stage * Genetic_Risk, data = df)
25 # Model 2: Cancer_Stage ? Obesity_Level
26 model_interact2 <- lm(Target_Severity_Score ~ Genetic_Risk + Cancer_Stage * Obesity_Level, data = df)
27 # Model 3: Full 3-way interaction
28 model_interact3 <- lm(Target_Severity_Score ~ Cancer_Stage * Obesity_Level * Genetic_Risk, data = df)
29
30 # Step 4: Compare Models Using ANOVA
31
32 anova(model_base, model_interact1)
33 anova(model_base, model_interact2)
34 anova(model_base, model_interact3)
35
36 #Step 5: Assumption Checks for Base Model
37
38 # 5.1 Linearity
39 crPlots(model_base)
40 # 5.2 Independence of Residuals
41 durbinwatsonTest(model_base)
42 # 5.3 Multicollinearity
43 vif(model_base)
44 # 5.4 Homoscedasticity
45 bptest(model_base)
46
47 # 5.5 Normality of Residuals
48 qqnorm(residuals(model_base))
49 qqline(residuals(model_base), col = "blue")
50 # 5.6 Influential Observations
51 plot(model_base, which = 4)
52
53 # Step 6: Robust Standard Errors (Optional)
54
55 coeftest(model_base, vcov = vcovHC(model_base, type = "HC3"))
```

5. Question 4 R code:

```
1 setwd("C:/Users/Almagd/OneDrive/Desktop/eco")
2 library(readxl)
3 cancer_data <- read_excel("C:/Users/Almagd/OneDrive/Desktop/cancer_modified (1)-2.xlsx")
4 View(cancer_data)
5 ##checking the impact of smoking , alcoholuse , obesity leveland gender on target severity score
6 reduced_model=lm (Target_Severity_Score ~ Smoking + Alcohol_Use + Obesity_Level + Gender , data = cancer_data)
7 summary(reduced_model)
8 ### checking if these effects vary across gender
9 full_model = lm (Target_Severity_Score ~ Smoking*Gender +Alcohol_Use*Gender + Obesity_Level*Gender , data = cancer_data)
10 summary(full_model)
11 ##choosing the best model
12 anova(reduced_model,full_model )
13 # Assumption checks
14 install.packages("car")
15 install.packages("lmtest")
16 library(car)
17 library(lmtest)
18 # 1. Linearity
19 crPlots(reduced_model)
20 # 2. Independence of errors
21 dwtest(reduced_model)
22 plot(residuals(reduced_model), type = "l", main = "Residuals in Observation Order")
23 abline(h = 0, col = "red")
24 # 3. Multicollinearity
25 # Variance Inflation Factors (VIF)
26 library(car)
27 # Run VIF on your linear model
28 vif(reduced_model) # Values >5-10 indicate problematic multicollinearity
29 # 4. Homoscedasticity
30 bptest(reduced_model)
31 plot(rstandard(reduced_model) ~ fitted(reduced_model),
32      main = "Residuals vs Fitted",
33      xlab = "Fitted Values",
34      ylab = "Standardized Residuals")
35 #Robust standard errors
36 library(sandwich)
37 library(lmtest)
38 coeftest(reduced_model, vcov = vcovHC(reduced_model, type = "HC1")) # Huber-white SEs
39 install.packages("estimatr")
40 #Load the package
41 library(estimatr)
42 # Fit model with robust SEs
43 reduced_model_robust <- lm_robust(Target_Severity_Score ~ Smoking + Alcohol_Use + Obesity_Level + Gender , data = cancer_data)
44 # Summary
45 summary(reduced_model_robust)
46 plot(fitted(reduced_model), residuals(reduced_model))
47 abline(h = 0, col = "red")
48 # 5. Normality (using QQ plot)
49 qqnorm(residuals(reduced_model))
50 qqline(residuals(reduced_model), col = "blue")
51 hist(rstandard(reduced_model))
52 # 6. Influential observations
53 plot(reduced_model, which = 4)
54 plot(reduced_model, which = 5)
55
```